

# UNIFORMITY, PRIME SUPPORT, AND REACHABLE LATTICES IN APPROACHES TO THE *abc* CONJECTURE

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**ABSTRACT.** We prove several estimates and structural statements relevant to the uniformity problem in *abc*. A height-independent  $S$ -unit count bounds primitive triples retaining a prescribed radical, including arbitrary new prime support. Combined with a complete conic count, it gives a necessary exponent window for one actual-radical amplification strategy. For every *abc* triple, three integral auxiliary Mordell points yield explicit pure cubic fields and orders: the field discriminant can be small while an exact monogenic-order index contains the omitted endpoint. Recent Thue estimates then give a relative height bound. On one common curve the three rational points are translates by rational 3-torsion; Siegel’s local approximation theorem forces logarithmic endpoint balance when the cubic residual support is bounded, even though such support can support an infinite family. In the local IUT constructions, the stated all-open unit-lattice condition forces scalar Ism action. This removes Ind2 from a fixed maximal-order module hull, while a cyclotomic trace invariant distinguishes specified native root-power inputs under Galois/Kummer transport. Their pure-orbit separation does not settle the subsequent hull or Ind3 comparison. We also prove effective finiteness of a specified Pell–Chebyshev family, with an absolute index bound  $p < 2^{59}$ , and give prime-support and endpoint tests for other proposed arguments. Selected arithmetic, lattice, and algebraic components are checked in Lean; external theorems and the remaining global interfaces are kept separate. Neither a proof nor a disproof of the standard *abc* conjecture is claimed.

## 1. THE TARGET AND THE ROLE OF THE RESULTS

Throughout, an *abc triple* means positive integers  $a, b, c$  such that  $a + b = c$  and  $a, b, c$  are pairwise coprime. Write

$$r = \text{rad}(abc), \quad R = \log r, \quad h = \log c.$$

Here  $\text{rad}(n)$  is the product of the distinct prime divisors of  $n$ . Pairwise coprimality is equivalent to  $\gcd(a, b, c) = 1$  under the additive relation. The target is the standard assertion

$$(1) \quad \forall \varepsilon > 0 \quad \exists C_\varepsilon \in \mathbb{R} \quad \forall (a, b, c) : \quad h \leq (1 + \varepsilon)R + C_\varepsilon.$$

Exponentiation gives the usual multiplicative constant. The repository’s `ABCConjecture` uses  $\log \max(a, b, c)$ , which equals  $h$  here. Its definition is not changed in this work.

The quantifier order in (1) is essential. Its negation requires one fixed positive  $\varepsilon$  and violations for every additive constant, hence at unbounded height. A finite set of high-quality triples cannot supply that negation: it is absorbed into  $C_\varepsilon$ . Likewise, a sparse exceptional-set bound does not imply that the exceptional set is finite.

Our results concern several specific interfaces. Section 2 uses finite prime support to test a proposed analytic construction. Section 3 bounds distinct outputs of two amplification constructions under explicit sufficient radical certificates. Section 5 records the cancellation that an endpoint approach must preserve. Section 6 supplies a local generator certificate relevant to possible-image constructions. Sections 8 and 9 examine effective arithmetic geometry and recent literature. In particular, Corollary 8.5 proves effective finiteness of a specified moving-index family. These components are partial theorems and applicability tests; none constructs the missing uniform bound in (1). Section 4 extends the amplification test from sufficient certificates to actual radicals. Section 7 uses the original admissible-arrow definitions and identifies

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a precise Ind2 hull equality. Section 10 supplies constructions for the full family of abc triples, together with their exact remaining endpoint and denominator costs. No priority over existing literature is claimed for consequences of the cited classical theorems.

The new statements below were proved mathematically before their corresponding Lean implementations. Separate research agents reviewed the arguments, source hypotheses, and formal scope; this internal review is not external journal peer review. Classical published theorems may be used in the paper, but are not silently represented as proved Lean declarations. The formal scope is specified in Section 11. The literature cutoff is August 30, 2026.

## 2. PRIME SUPPORT IN SHORT INTERVALS

Let  $\omega(n)$  be the number of distinct prime divisors of  $n$ , and let  $P^+(n)$  be its greatest prime divisor, with  $P^+(1) = 1$ . For integers  $N, Y \geq 1$  and  $w \geq 0$ , define

$$\mathcal{A}(N, Y, w) = \{1 \leq n \leq N : P^+(n) \leq Y, \omega(n) \leq w\}.$$

**Theorem 2.1** (Finite prime-power encoding). *With  $L = \lfloor \log_2 N \rfloor$ , one has*

$$(2) \quad |\mathcal{A}(N, Y, w)| \leq (1 + YL)^w.$$

*Proof.* Use the alphabet

$$\mathcal{Q} = \{1\} \cup \{q^e : 1 \leq q \leq Y, 1 \leq e \leq L\}.$$

It has at most  $1 + YL$  members; the bases need not be prime. For  $n \in \mathcal{A}(N, Y, w)$ , write its actual prime factorization as  $n = \prod_{j=1}^t p_j^{e_j}$ , with  $p_1 < \dots < p_t$  and  $t \leq w$ . Since  $2^{e_j} \leq p_j^{e_j} \leq n \leq N$ , each  $e_j \leq L$ . Thus every factor belongs to  $\mathcal{Q}$ . Append  $w - t$  copies of 1 to obtain a word of length  $w$ . Its product recovers  $n$ , so different integers have different encoded words. Counting all words proves (2). The empty factorization covers  $n = 1$  and  $w = 0$ ;  $N = 1$  is also included.  $\square$

**Corollary 2.2** (A finite first-moment bound). *Let  $S$  be a finite set of distinct positive  $Y$ -smooth integers at most  $N$ , and put  $B = (1 + Y \lfloor \log_2 N \rfloor)^w$ . Then*

$$(3) \quad |\{n \in S : \omega(n) \leq w\}| \leq B,$$

$$(4) \quad \sum_{n \in S} \omega(n) \geq (w + 1) \max\{|S| - B, 0\}.$$

*Proof.* The first set is contained in  $\mathcal{A}(N, Y, w)$ . Its complement in  $S$  has at least  $\max\{|S| - B, 0\}$  members, each with at least  $w + 1$  distinct prime factors. Summation proves the second assertion.  $\square$

Suppose now that  $y = y(x) \geq 2$  satisfies

$$(5) \quad \log \log x = o(\log y), \quad \log y = o(\log x), \quad u = \frac{\log x}{\log y}.$$

For every fixed  $\kappa > 0$ , Theorem 2.1, with  $N = \lfloor 2x \rfloor$ ,  $Y = \lfloor y \rfloor$  and  $w = \lfloor \kappa u \rfloor$ , gives

$$(6) \quad \#\{n \leq 2x : P^+(n) \leq y, \omega(n) \leq \kappa u\} \leq x^{\kappa + o(1)}.$$

Indeed,

$$w \log(1 + YL) \leq \kappa \frac{\log x}{\log y} (\log y + O(\log \log x)) = \kappa \log x + o(\log x).$$

More generally the upper bound is  $x^{o(1)}$  whenever  $w(\log y + \log \log x) = o(\log x)$ .

**2.1. The external analytic input.** Write  $\Psi(x, y) = \#\{1 \leq n \leq x : P^+(n) \leq y\}$ . Younis's unconditional Theorem 1.1 [27], p. 2, states that for each fixed  $17/30 < \theta \leq 1$  there is  $C_Y(\theta) > 0$  such that

$$(7) \quad \frac{\Psi(x+h, y) - \Psi(x, y)}{h} = \frac{\Psi(x, y)}{x} \left( 1 + O_\theta \left( \frac{\log(u+1)}{\log y} \right) \right)$$

uniformly for

$$x^\theta \leq h \leq x, \quad \exp\left(C_Y(\theta)(\log x)^{2/3}(\log \log x)^{4/3}\right) \leq y \leq 2x.$$

Fix

$$(8) \quad \frac{17}{30} < \theta < 1, \quad C \geq C_Y(\theta), \quad h = x^\theta, \quad y = \exp\left(C(\log x)^{2/3}(\log \log x)^{4/3}\right).$$

The lower bound on  $C$  is part of the hypotheses, not a disposable constant. Put

$$S_x = \{n \in (x, x+h] : P^+(n) \leq y\}, \quad M_x = |S_x|.$$

The classical long-interval estimate recalled in [7, equations (1.8)–(1.9), p. 415] yields  $\Psi(x, y) = x\rho(u)(1+o(1))$  here. Its range  $u \leq (\log y)^{3/5-\varepsilon}$  is satisfied, for example with  $\varepsilon = 1/20$ : the power of  $\log x$  on the right is  $11/30$ , larger than the power  $1/3$  in  $u$ . Also  $-\log \rho(u) = u \log u(1+o(1)) = o(\log x)$ . Consequently (7) gives

$$(9) \quad M_x = h\rho(u)(1+o(1)) = x^{\theta-o(1)}.$$

Only this mass estimate is needed below; no short-interval moment formula is assumed.

**Theorem 2.3** (A lower tail and a first-moment obstruction). *For (8) and every fixed  $0 < \kappa < \theta$ ,*

$$(10) \quad \frac{\#\{n \in S_x : \omega(n) \leq \kappa u\}}{M_x} \leq x^{\kappa-\theta+o(1)} \rightarrow 0.$$

*In particular,*

$$(11) \quad \liminf_{x \rightarrow \infty} \frac{1}{uM_x} \sum_{n \in S_x} \omega(n) \geq \theta.$$

*For every fixed  $A > 0$ , the proportion of  $n \in S_x$  with  $\omega(n) \leq (\log \log x)^A$  tends to zero.*

*Proof.* Every member of  $S_x$  is at most  $2x$ . Divide (6) by (9) to obtain (10). If its numerator is  $B_x$ , then

$$\sum_{n \in S_x} \omega(n) \geq \kappa u(M_x - B_x).$$

The lower limit after division by  $uM_x$  is at least each fixed  $\kappa < \theta$ , and hence at least  $\theta$ . This argument does not require a variable  $\kappa$  or a uniform error term in that variable. Finally,

$$(\log \log x)^A (\log y + \log \log x) = o(\log x).$$

The last observation after (6) bounds the corresponding population by  $x^{o(1)}$ . Divide again by (9).  $\square$

**Corollary 2.4** (A false intermediate assertion in a claimed disproof). *Lemma 4.2 of Carella [6], p. 8, equation (4.5), is false in its stated range. The same holds for the second-moment assertion in Lemma 4.4, p. 9, and the relative exceptional estimate (4.24), p. 11.*

*Proof.* Choose  $\theta = 3/5$  and a fixed  $C \geq C_Y(3/5)$ . Then  $h > x^{7/12}$ ,  $y < h < x$ , and  $y = o(h)$ , exactly as required in the first lemma. Its proposed formula would give

$$\frac{1}{M_x} \sum_{n \in S_x} \omega(n) = O(\log \log y).$$

However,

$$\frac{u}{\log \log y} \asymp \frac{(\log x)^{1/3}}{(\log \log x)^{7/3}} \rightarrow \infty,$$

contradicting (11). Changing the left endpoint of the interval from open to closed adds at most one integer; its contribution is at most  $\log_2(2x)$  and is negligible relative to  $M_x$ . The proposed second moment is  $O(M_x(\log \log y)^2)$ , whereas (10) gives a lower bound  $\kappa^2 u^2 M_x(1 - o(1))$  for any fixed  $0 < \kappa < 3/5$ . Finally the threshold in (4.24) is bounded by a fixed power of  $\log \log x$ . Theorem 2.3 shows that the proportion above it tends to one, although the claimed relative upper bound tends to zero.  $\square$

*Remark 2.5.* This refutes those intermediate assertions, not  $abc$  and not the possibility of an exceptionally sparse counterexample family. In particular a set of size  $x^{o(1)}$  can still be infinite. It is also important to distinguish  $o(h)$  from  $o(M_x)$ : the whole smooth set already has size  $o(h)$ .

**2.2. Two further transfer tests.** For a fixed integer  $k \geq 4$ , the intervals

$$(p^k, p^k + (p^k)^{3/5}], \quad X < p^k \leq 2X,$$

are pairwise disjoint for large  $X$ . Indeed, successive distinct integer bases give gaps at least  $X^{1-1/k}$ , larger than  $(2X)^{3/5}$ . The finite encoding, using the largest thresholds in this range, shows that at most  $X^{o(1)}$  centres can contain a smooth candidate with  $\omega(n) \leq 2 \log \log(p^k)$  at the scale (8). The prime number theorem gives  $X^{1/k+o(1)}$  centres, so their proportion is at most  $X^{-1/k+o(1)}$ . This rules out a positive-proportion selection of this particular kind, while leaving a sparse selection unresolved.

Recent work on exceptional sets [3, 4] and on the existence of smooth neighbours [10] addresses different quantifiers. A useful general check is the following. If a counting theorem gives  $N(T) \ll T^{\eta+o(1)}$ , then a seed of height  $H$  contradicts it only when it produces more than  $H^{\beta\eta+o(1)}$  distinct qualifying outputs of height at most  $O(H^\beta)$ . Repeated encodings of one output do not count. None of the existence results just cited supplies the required uniform low-radical neighbours or such an amplification theorem.

### 3. TWO EXPLICIT AMPLIFICATION FAMILIES

Let  $N_\mu(T)$  count positive primitive triples  $A + B = C \leq T$  with  $\text{rad}(ABC) < C^\mu$ , where  $0 < \mu < 1$ . The current estimates [3, Proposition 1.1 and Theorems 1.2–1.3] give

$$(12) \quad N_\mu(T) \ll_{\mu,\delta} T^{F(\mu)+\delta}, \quad F(\mu) = \min \left\{ \frac{2\mu}{3}, \frac{23\mu+3}{40}, \frac{3}{5} \right\}, \quad \delta > 0.$$

We test two actual constructions from a primitive seed  $a + b = c$ . An amplifier contradicting (12) at output height  $T = c^K$ , for fixed  $K > 0$ , would need more than  $c^{KF(\mu)+o(1)}$  distinct qualifying outputs. Our conclusions concern explicit radical certificates, rather than the entire exceptional set.

**3.1. Inherited congruence moduli.** Fix pairwise coprime positive integers  $U, V, W$ , put  $M = UVW$ , and let  $R_M = \text{rad}(M)$ . Outputs with  $U \mid A, V \mid B, W \mid C$  satisfy

$$\text{rad}(ABC) \leq \frac{R_M ABC}{M}.$$

Call an output *size-certified* if the right side is at most  $C^\mu$ .

**Proposition 3.1.** *For  $T \geq 1$ , there are at most  $1 + \lfloor \log_2 T \rfloor$  size-certified primitive outputs with  $C \leq T$ , for  $0 < \mu \leq 1$ . Allowing every inherited template  $U \mid a, V \mid b, W \mid c$  gives at most*

$$\tau(abc)(1 + \lfloor \log_2 T \rfloor)$$

*distinct outputs. For fixed  $K$  and  $T \leq c^K$  this is  $c^{o(1)}$ .*

*Proof.* Suppose first  $M \geq 2$ . Certification implies  $R_M AB \leq M$ , and  $R_M \geq 2$ . Consider two outputs in the strip  $L \leq A, A' < 2L$ . Their determinant  $\Delta = AB' - A'B$  is divisible by  $UV$  and also by  $W$ , since  $\Delta = AC' - A'C$ . Thus  $M \mid \Delta$ . The bounds  $B' \leq M/(R_M A')$  and  $B \leq M/(R_M A)$  give

$$0 < AB', A'B < 2M/R_M \leq M.$$

Consequently  $|\Delta| < M$ , so  $\Delta = 0$ . Coprimality implies  $A \mid A'$  and  $A' \mid A$ , hence equality of all coordinates. There is at most one output in each dyadic strip. The strict strip endpoint also handles  $R_M = 2$ . If  $M = 1$ , certification forces  $AB \leq 1$ ; only  $(1, 1, 2)$  can remain, and only when the exponent permits it.

The number of inherited templates is  $\tau(a)\tau(b)\tau(c) = \tau(abc)$ . A union bound proves the second claim. For every fixed  $\eta > 0$ , the elementary estimate  $\tau(n) \ll_\eta n^\eta$  follows from  $e + 1 \leq 2^e \leq p^{\eta e}$  for sufficiently large primes  $p$  and bounded multiplicative costs at the finitely many smaller primes. Since  $abc \leq c^3$ , this proves the last claim.  $\square$

Outputs whose remaining factors have additional powers or smaller actual radicals are not ruled out by this proposition. A count of raw CRT solutions cannot replace the count of distinct certified primitive triples.

**3.2. Square completion on a conic.** Put  $P = abc$  and  $R_P = \text{rad}(P)$ . Positive integer points on

$$(13) \quad ax^2 + by^2 = cz^2$$

give candidate outputs  $(ax^2, by^2, cz^2)$ . Retain only primitive outputs. Then  $\gcd(x, y, z) = 1$ , and the positive coordinates are recovered uniquely from the output. Their radical is at most  $R_Pxyz$ .

**Lemma 3.2.** *For coprime integers  $m, n$ , not both zero, put*

$$Q = an^2 + bm^2, \quad J = an + bm, \quad X = Q - 2nJ, \quad Y = Q - 2mJ,$$

*and set*

$$D_0 = \gcd(a, m) \gcd(b, n) \gcd(c, m - n), \quad G = \gcd(X, Y, Q) > 0.$$

*Then*

$$aX^2 + bY^2 = cQ^2, \quad \gcd(Q, J) = D_0, \quad G = \gcd(Q, 2J), \quad D_0 \mid G \mid 2D_0.$$

*Every positive primitive point of (13) is represented by  $(X/G, Y/G, Q/G)$  for some such pair. Passing to absolute values only overcounts outputs, and  $(m, n)$  and  $(-m, -n)$  give the same point.*

*Proof.* Expansion proves the conic identity. Coprimality of  $m, n$  gives  $\gcd(Q, 2nJ, 2mJ) = \gcd(Q, 2J)$ . Here is a valuation check of the remaining gcd, including the prime two. If a prime divides both  $Q, J$ , the identities

$$aQ = J^2 - 2bmJ + bcm^2, \quad bQ = J^2 - 2anJ + acn^2$$

show that it divides  $abc$ . At a prime dividing  $a$ , the common valuation is  $\min(v_p(a), v_p(m))$ : compare the terms of  $J = an + bm$  and  $Q = an^2 + bm^2$ , using that  $b, n$  are units whenever  $p \mid m$ . The case  $p \mid b$  is identical with the roles exchanged. At  $p \mid c$ , set  $e = v_p(c)$ ,  $r = v_p(n - m)$ . If  $r = 0$  the common valuation is zero. If  $r > 0$ , both coordinates are units and  $J = a(n - m) + cm$  has valuation  $\min(e, r)$  unless  $e = r$ . In that exceptional case either  $v_p(J) = e$ , or

$$Q = 2mJ - cm^2 + a(n - m)^2$$

has valuation exactly  $e$ . Thus the common valuation is always  $\min(e, r)$ . This argument does not divide by two. It proves the gcd formula and hence the assertions about  $G$ .

In the chart  $z = 1$ , a line through  $(1, 1)$  has equation  $(x, y) = (1, 1) + t(n, m)$ . Its intersection parameters satisfy  $t(2J + tQ) = 0$ , giving the displayed second point. Every other rational point determines such a rational direction, which may be chosen primitive. The base point itself comes from the tangent direction  $(m, n) = (-a, b)$ . Dividing homogeneous coordinates by their gcd gives every primitive point.  $\square$

**Theorem 3.3.** *For every  $T > 0$ , the number of positive primitive square-completion outputs with  $cz^2 \leq T$  is at most*

$$(14) \quad \tau(P) \left(1 + 4\pi\sqrt{T/P}\right).$$

The subfamily certified by  $R_Pxyz \leq (cz^2)^\mu$ , for  $0 < \mu < 1$ , has size at most

$$(15) \quad \tau(P) \left( 1 + \frac{4\pi}{\sqrt{c}} T^{\mu/2} \right).$$

*Proof.* Fix the three divisors in  $D_0$  from Lemma 3.2. The conditions

$$d_a \mid m, \quad d_b \mid n, \quad d_c \mid m - n$$

define a lattice in  $\mathbb{Z}^2$  of index  $D_0 = d_a d_b d_c$ . Indeed the first two have index  $d_a d_b$ , and the third residue map is surjective there because these three divisors are pairwise coprime. The height bound and  $G \leq 2D_0$  imply

$$an^2 + bm^2 = Q \leq 2D_0 \sqrt{T/c}.$$

This ellipse has area  $V = 2\pi D_0 \sqrt{T/P}$ .

A centred ellipse of area  $V$  contains at most  $1 + 2V/D_0$  primitive integer directions in a lattice of index  $D_0$ , modulo sign. To see this, select representatives in a fixed half-plane and order them by angle. There is at most one primitive vector on each positive ray. Consecutive distinct directions have nonzero determinant divisible by  $D_0$ , so their triangles with the origin have area at least  $D_0/2$ . These triangles have disjoint interiors and lie in the ellipse. The cases of zero or one direction satisfy the bound as well. There are  $\tau(P)$  divisor templates, proving (14).

For a certified output write  $C = cz^2$ . Equation (13) implies  $\max(x, y) \geq z$ , so  $xyz \geq z^2$ . Hence

$$R_P C / c \leq R_P xyz \leq C^\mu, \quad C^{1-\mu} \leq c / R_P.$$

All certified heights are at most  $U = \min\{T, (c/R_P)^{1/(1-\mu)}\}$ . Moreover,  $ab \geq c - 1 \geq c/2$  and  $R_P \geq 2$ , giving  $R_P P \geq c^2$ . Therefore

$$\sqrt{U/P} = U^{\mu/2} \sqrt{U^{1-\mu}/P} \leq T^{\mu/2} \sqrt{c/(R_P P)} \leq c^{-1/2} T^{\mu/2}.$$

Apply (14) at height  $U$  to obtain (15).  $\square$

For fixed  $K > 0$  and  $T = c^K$ , (15) is  $c^{\max\{0, K\mu/2 - 1/2\} + o(1)}$ . Every term defining  $F(\mu)$  is strictly greater than  $\mu/2$ , so this family cannot exceed the exponent required to contradict (12). The conclusion is limited to the certificate. For example, the seed  $(1, 8, 9)$  and point  $(7, 2, 3)$  give  $(49, 32, 81)$ , whose radical is 42, while  $R_P xyz = 252$ . Since  $42^2 < 3^7$ ,  $42 < 81^{9/10}$ , although the certificate fails even with exponent one. This example proves that actual exceptional descendants cannot be identified with the certified subfamily; it is not a disproof of abc.

#### 4. ACTUAL RADICAL BUDGETS AND A NECESSARY AMPLIFICATION WINDOW

Here we count actual prime support, allowing arbitrary new primes and their repeated powers. Fix a positive primitive seed  $a + b = c$  and put  $P = abc$ . *Only in this section*,  $R = \text{rad}(P)$  denotes the integer radical, rather than its logarithm as in the introductory notation. Thus  $c \geq 2$  and  $R \leq P \leq c^3$ . All output triples below are ordered.

**Theorem 4.1** (A height-independent radical budget). *Let  $R \geq 1$  be a squarefree integer and  $Y > 0$  a real number. The number  $\mathcal{N}(R, Y)$  of all positive primitive triples  $A + B = C$  with  $R \mid \text{rad}(ABC)$  and  $\text{rad}(ABC) \leq Y$  is zero if  $Y < R$ . If  $Y \geq R$ , then, with  $X = Y/R \geq 1$ ,*

$$(16) \quad \mathcal{N}(R, Y) \leq 905 \cdot 45^{\omega(R)} X(1 + \log X)^{44}.$$

*There is no restriction on the output height  $C$ .*

*Proof.* For each output,  $D = \text{rad}(ABC)/R$  is a squarefree integer with  $D \leq X$  and  $\gcd(D, R) = 1$ . Fix  $D$  and take  $S$  to consist of the archimedean place of  $\mathbb{Q}$  and the primes dividing  $RD$ . The pair  $(A/C, B/C)$  solves  $u + v = 1$  in  $S$ -units. This association is injective:  $A/C$  is reduced because  $\gcd(A, C) = 1$ , so it recovers  $A, C$  and then  $B$ .

Theorem 1.1 of [8] bounds the number of solutions over a degree- $m$  number field by  $(3.1 + 5(3.4)^m)45^{|S|}$ . Here  $m = 1$  and  $|S| = 1 + \omega(R) + \omega(D)$ , including the infinite place. The number



for this  $D$  is consequently at most  $904.545^{\omega(R)+\omega(D)}$ , which is smaller than the same expression with coefficient 905.

For an integer  $M \geq 1$ , let  $d_M(n)$  count positive ordered  $M$ -tuples of product  $n$ . Prime factorization gives

$$M^{\omega(n)} \leq d_M(n) = \prod_{p^e \parallel n} \binom{e+M-1}{M-1}.$$

Fixing the first  $M-1$  entries of a tuple, and then dropping their product restriction, yields for every real  $X \geq 1$

$$\begin{aligned} \sum_{n \leq X} d_M(n) &\leq X \sum_{n_1 \cdots n_{M-1} \leq X} \frac{1}{n_1 \cdots n_{M-1}} \\ &\leq X \left( \sum_{1 \leq n \leq X} \frac{1}{n} \right)^{M-1} \leq X(1 + \log X)^{M-1}. \end{aligned}$$

The empty tuple covers  $M=1$ , and every sum equals one when  $X=1$ . Sum the fixed- $D$  estimate, enlarge the sum by discarding coprimality and squarefreeness, and use  $M=45$ . This proves the bound and finiteness. The case  $Y < R$  follows from divisibility of positive integers.  $\square$

**Corollary 4.2** (Uniformity for a seed). *Fix  $K > 0$ ,  $0 < \mu < 1$ , and  $\varepsilon > 0$ . The number of outputs with*

$$R \mid \text{rad}(ABC), \quad \text{rad}(ABC) \leq C^\mu, \quad C \leq T \leq c^K$$

*is zero if  $T^\mu < R$ , and otherwise is*

$$(17) \quad \ll_{\varepsilon, K, \mu} c^\varepsilon \frac{T^\mu}{R}.$$

*The implied constant is independent of the seed and of  $T$ .*

*Proof.* Apply Theorem 4.1 with  $Y = T^\mu$ . For any fixed  $\eta > 0$ , separating the finitely many primes  $p < 45^{1/\eta}$  gives  $45^{\omega(R)} \ll_\eta R^\eta \leq c^{3\eta}$ . Choose  $\eta = \varepsilon/6$  before varying the seed. Since  $\log(T^\mu/R) \leq K\mu \log c$ , the remaining logarithmic factor is  $\mathcal{O}_{\varepsilon, K, \mu}(c^{\varepsilon/2})$ . This proves the assertion.  $\square$

To record exactly what repeated primes save, for positive integers  $P, n$  set

$$D_P(n) = \prod_{\substack{p \mid n \\ p \nmid P}} p, \quad E_P(n) = n/D_P(n) \in \mathbb{N}_{>0}.$$

The factor  $D_P(n)$  divides  $n$ , is squarefree, and is coprime to  $R$ . In  $E_P(n)$ , an old prime contributes its entire exponent, and a new prime of exponent  $e$  contributes exponent  $e-1$ .

**Proposition 4.3** (Exact excess accounting). *Let  $x, y, z$  be positive integers satisfying  $ax^2 + by^2 = cz^2$ , and put  $(A, B, C) = (ax^2, by^2, cz^2)$ . Then*

$$(18) \quad \text{rad}(ABC) = RD_P(xyz), \quad \text{rad}(ABC)E_P(xyz) = Rxyz.$$

*In particular,  $\text{rad}(ABC) \leq C^\mu$  implies*

$$(19) \quad E_P(xyz) \geq (R/c)C^{1-\mu}.$$

*These statements do not require the output to be primitive.*

*Proof.* The prime divisors of  $Pn^2$  are exactly those of  $Pn$ , so  $\text{rad}(Pn^2) = RD_P(n)$ . Use  $ABC = P(xyz)^2$  to obtain (18). The conic equation makes  $z^2$  a positive weighted average of  $x^2, y^2$ . Thus  $\max(x, y) \geq z$ , and  $x, y \geq 1$  give  $xyz \geq z^2 = C/c$ . Consequently  $RC/c \leq Rxyz = \text{rad}(ABC)E_P(xyz) \leq C^\mu E_P(xyz)$ .  $\square$

**Theorem 4.4** (A necessary amplification window). *Put  $\rho = \log P / \log c$  and  $\sigma = \log R / \log c$ . Fix  $K > 0$  and  $0 < \mu < 1$ . Consider all positive integer square completions whose unscaled output  $(ax^2, by^2, cz^2)$  is primitive and satisfies  $C \leq c^K$  and  $\text{rad}(ABC) \leq C^\mu$ . If nonempty, this family has size*

$$(20) \quad \ll_{\varepsilon, K, \mu} c^{\min\{\max(0, (K-\rho)/2), K\mu-\sigma\}+\varepsilon} \quad (\varepsilon > 0).$$

For the exponent before  $\varepsilon$  to exceed  $KF(\mu)$  from (12), it is necessary that

$$(21) \quad K > \rho + 4\sigma, \quad \frac{3\sigma}{K} < \mu < \frac{3}{4} \left(1 - \frac{\rho}{K}\right) < \frac{3}{4}.$$

All implied constants are uniform in the seed for fixed  $K, \mu, \varepsilon$ .

*Proof.* Integer square completion retains every prime of  $P$ , so Corollary 4.2 gives the exponent  $K\mu - \sigma$ ; nonemptiness makes it nonnegative. The complete fibre bound (14) in Theorem 3.3, together with  $\tau(P) \ll_\eta P^\eta$  and  $P \leq c^3$ , gives  $\max(0, (K - \rho)/2)$ . Taking the smaller bound proves (20). The elementary divisor estimate used here follows by splitting primes at a fixed threshold; for large primes,  $e + 1 \leq p^{\eta e}$ , and for each remaining prime the quotient has a finite supremum over  $e \geq 0$ .

The BBLT exponent is  $F(\mu) = \min\{2\mu/3, (23\mu + 3)/40, 3/5\}$  [3]. If  $\mu \geq 3/4$ , all three entries are at least  $1/2$ , whereas  $\max(0, (K - \rho)/2) < K/2$  because  $K, \rho > 0$ . Thus the desired strict exponent gain is impossible in this range, even for the complete fibre. If  $0 < \mu < 3/4$ , then  $F(\mu) = 2\mu/3$ . Both exponents in (20) must exceed  $2K\mu/3$ , forcing

$$K\mu - \sigma > 2K\mu/3, \quad (K - \rho)/2 > 2K\mu/3.$$

These inequalities give the two bounds on  $\mu$  in (21); their interval is nonempty only if  $K > \rho + 4\sigma$ .  $\square$

In particular, along seeds satisfying  $K \leq \rho + 4\sigma$ , this family cannot supply a uniform lower count  $c^\alpha$  with a fixed gap  $\alpha > KF(\mu)$ . Choose  $\varepsilon$  in (20) smaller than that gap to see this directly. The conclusion is for fixed  $K, \mu$ ; it asserts no uniformity for growing  $K$ . An exceptional seed bounds  $\sigma$  from above, and that must not be substituted as a lower bound in the counting estimate. The interval (21) remains open: no sufficient lower bound for actual excesses or distinct outputs has been proved here.

Full support retention also needs its stated domain. The primitive seed  $(49, 576, 625)$ , of radical 210, has the rational completion  $(x, y, z) = (3/7, 1/6, 1/5)$  with primitive output  $(9, 16, 25)$  of radical 30. Equivalently, the integer point  $(90, 35, 42)$  loses the prime 7 after its output is divided by the common factor 44100. Thus arbitrary projective normalization is not covered by the full- $R$  estimate. If the output itself retains the rational squareclasses  $ax^2, by^2, cz^2$ , only primes of odd valuation in  $P$  are automatically retained: their corresponding output valuations are nonnegative odd integers.

A separate attempt uses the three quadratic tripod identities

$$((a - b)^2, 4ab, c^2), \quad (a^2, 4bc, (b + c)^2), \quad (b^2, 4ac, (a + c)^2),$$

with  $a \neq b$  and each row divided by its actual common divisor. That divisor is 1 or 4: for coprime  $u, v$ , the entries  $(u - v)^2, 4uv, (u + v)^2$  have no common odd prime, and their gcd is 4 exactly when both  $u, v$  are odd. Their primitive radicals are exactly

$$R \text{rad}_{\text{odd}}(|a - b|), \quad R \text{rad}_{\text{odd}}(a + 2b), \quad R \text{rad}_{\text{odd}}(2a + b),$$

where  $\text{rad}_{\text{odd}}$  omits the prime 2. Every new odd prime in the displayed factor is coprime to  $P$ , while all seed odd primes survive normalization and every primitive positive triple contains 2. Thus all three branches, and their positive iterates, retain  $R$ .

Nevertheless, no branch is guaranteed to preserve quality  $q = \log C / \log \text{rad}(ABC)$ . The seed  $(1, 8, 9)$  gives  $(49, 32, 81)$ ,  $(1, 288, 289)$ , and  $(16, 9, 25)$ , with radicals 42, 102, 30. The first quality is smaller than  $\log 9 / \log 6$  because  $42 > 36$ ; the last is less than one. For the middle one,

$$\frac{\log 289}{\log 102} < \frac{38}{31} < \frac{\log 9}{\log 6}, \quad 17^{24} < 6^{38}, \quad 2^{38} < 3^{24},$$



where the integer comparisons prove the two logarithmic inequalities. This refutes only the proposed universal quality-preserving selection among these three branches. Neither finite example is an *abc* disproof, and more general maps and the remaining window are not excluded.

## 5. THE COUPLED SIGNED PRIME-EXPONENT DEFECT

For  $n \geq 1$  put

$$E_1(n) = \sum_{\substack{p|n \\ v_p(n)=1}} \log p, \quad E_3(n) = \sum_{p|n} (v_p(n) - 2)_+ \log p.$$

Write  $m = \min(a, b)$  and  $M = \max(a, b)$  for an *abc* triple.

**Proposition 5.1** (Exact signed identity). *For every positive integer  $n$ ,*

$$(22) \quad \delta_2(n) := \log n - 2 \log \text{rad}(n) = E_3(n) - E_1(n).$$

*The coupled defect satisfies*

$$(23) \quad \begin{aligned} \Delta &:= \log(Mc) - 2R \\ &= E_3(M) + E_3(c) - E_1(M) - E_1(c) - 2 \log \text{rad}(m), \end{aligned}$$

$$(24) \quad 2h - 2R - \log 2 \leq \Delta \leq 2h - 2R.$$

*Consequently *abc* is equivalent to*

$$(25) \quad \forall \varepsilon > 0 \quad \exists K_\varepsilon \quad \forall (a, b, c) : \quad E_3(M) + E_3(c) \leq E_1(M) + E_1(c) + 2 \log \text{rad}(m) + 2\varepsilon R + K_\varepsilon.$$

*Proof.* For every positive exponent  $e$ ,  $e - 2 = (e - 2)_+ - \mathbf{1}_{e=1}$ . Multiply by  $\log p$  and sum over the actual prime factorization to obtain (22). Pairwise coprimality gives  $r = \text{rad}(m) \text{rad}(M) \text{rad}(c)$ , which proves (23). Since  $c/2 \leq M \leq c$ , taking logarithms of  $c^2/2 \leq Mc \leq c^2$  gives (24). An *abc* constant  $C_\varepsilon$  supplies  $K_\varepsilon = 2C_\varepsilon$ . Conversely (25) and the lower corridor give  $C_\varepsilon = (K_\varepsilon + \log 2)/2$ .  $\square$

The equivalence does not prove (25); it identifies the entire missing estimate without losing the negative exponent-one terms. Replacing the coupled inequality by separate endpoint bounds fails.

**Proposition 5.2** (An unbounded separate-endpoint obstruction). *For every  $0 \leq \rho < 1$  and  $K \in \mathbb{R}$ , there is an *abc* triple with  $m = 1$  and*

$$\delta_2(M) > \log \text{rad}(m) + \rho R + K.$$

*Proof.* Take  $(a, b, c) = (1, 2^N, 2^N + 1)$  with  $N \geq 1$ . Then  $m = 1$ ,  $M = 2^N$ , and  $\delta_2(M) = (N - 2) \log 2$ . Without making any assumption about the odd neighbour,

$$r \leq 2(2^N + 1) \leq 2^{N+2}, \quad R \leq (N + 2) \log 2.$$

Therefore

$$\delta_2(M) - \rho R \geq ((1 - \rho)N - 2 - 2\rho) \log 2 \longrightarrow \infty.$$

This proves the claim. The bound uses only  $\text{rad}(n) \leq n$  and  $\text{rad}(2^N) = 2$ .  $\square$

This argument neither proves nor assumes that all odd neighbours have large radical. It only rejects a false stronger estimate that discards their possible contribution. The coupled route remains open.

## 6. FINITE CERTIFICATES FOR REACHABLE LOCAL LATTICES

Let  $K$  be a nonarchimedean local field,  $\mathcal{O}$  its valuation ring,  $\pi$  a uniformizer, and  $k = \mathcal{O}/\pi\mathcal{O}$  its residue field, with  $|k| = q > 1$  and  $v(\pi) = 1$ . Fix  $N \geq 1$ , write  $L = \mathcal{O}^N$ , and normalize additive Haar measure on  $K^N$  by  $\mu(L) = 1$ . For any set  $\mathcal{R} \subset L$ , let

$$H(\mathcal{R}) = \overline{\text{span}_{\mathcal{O}}(\mathcal{R})}.$$

If nonsingular  $N$ -tuples exist in  $\mathcal{R}$ , define

$$d(\mathcal{R}) = \min_{\substack{r_1, \dots, r_N \in \mathcal{R} \\ \det[r_1 \ \dots \ r_N] \neq 0}} v(\det[r_1 \ \dots \ r_N]);$$

otherwise set  $d(\mathcal{R}) = \infty$ .

**Theorem 6.1** (A finite witness for the exact hull measure). *With  $q^{-\infty} = 0$ , one has*

$$\mu(H(\mathcal{R})) = q^{-d(\mathcal{R})}.$$

*If  $d(\mathcal{R}) < \infty$ , some matrix  $A$  whose columns belong to  $\mathcal{R}$  satisfies*

$$v(\det A) = d(\mathcal{R}), \quad H(\mathcal{R}) = A\mathcal{O}^N.$$

*Proof.* In the full-rank case the nonzero determinant valuations are nonnegative integers, so choose  $A$  attaining their minimum  $d$ . For  $r \in \mathcal{R}$ , Cramer's rule gives

$$(A^{-1}r)_i = \frac{\det A_i(r)}{\det A},$$

where  $A_i(r)$  replaces column  $i$  by  $r$ . Its columns still belong to  $\mathcal{R}$ . Every nonzero numerator has valuation at least  $d$ ; a zero numerator causes no difficulty. Thus  $A^{-1}r \in \mathcal{O}^N$  and  $\mathcal{R} \subset A\mathcal{O}^N$ . Since the columns of  $A$  lie in  $\mathcal{R}$ , their integral spans are equal. The set  $A\mathcal{O}^N$  is compact, hence closed.

Integral row and column operations give  $UAV = \text{diag}(\pi^{a_1}, \dots, \pi^{a_N})$ , with  $U, V \in \text{GL}_N(\mathcal{O})$ ,  $a_i \geq 0$ , and  $\sum a_i = d$ . One may obtain this diagonalization by repeatedly moving an entry of least valuation to the upper left and clearing its row and column by integral operations. The quotient  $\mathcal{O}^N/A\mathcal{O}^N$  has  $q^d$  elements. Its equal-measure cosets partition  $L$ , proving  $\mu(A\mathcal{O}^N) = q^{-d}$ .

In the rank-deficient case a nonzero  $K$ -linear functional annihilates  $\mathcal{R}$ . Scale its coefficients to lie in  $\mathcal{O}$  with at least one unit coefficient. It then defines a surjection  $\ell : L \rightarrow \mathcal{O}$  whose closed kernel contains  $H(\mathcal{R})$ . For every  $t \geq 1$ ,  $\ell^{-1}(\pi^t\mathcal{O})$  has index  $q^t$  in  $L$ . Hence  $0 \leq \mu(H(\mathcal{R})) \leq q^{-t}$  for all  $t$ , proving measure zero.  $\square$

**Corollary 6.2.** *The equality  $H(\mathcal{R}) = L$  holds if and only if the reductions of  $\mathcal{R}$  span  $k^N$ , equivalently if some reachable column matrix has unit determinant. More generally, any measurable  $\mathcal{O}$ -submodule containing all columns of a reachable matrix  $A$  with  $v(\det A) = d$  has measure at least  $q^{-d}$ .*

*Proof.* A determinant is a unit exactly when its reduction is nonzero; a spanning set of  $k^N$  contains a basis. Apply Theorem 6.1. For the last claim, the target contains  $A\mathcal{O}^N$  and measure is monotone. The target need not be contained in  $L$ .  $\square$

**6.1. Independent and simultaneous actions.** Let  $L_j = \mathcal{O}^{d_j}$ , with  $d_j \geq 1$ , and let  $T = \bigotimes_{j=1}^s L_j$ , where  $s \geq 1$  and  $N = \prod_j d_j$ . For nonzero  $x_j \in L_j$ , write  $x_j = \pi^{m_j} u_j$  with  $u_j$  primitive.

**Proposition 6.3.** *Under the independent action of  $\prod_j \text{GL}_{d_j}(\mathcal{O})$ , the orbit of  $\bigotimes_j x_j$  spans  $\pi^{\sum_j m_j} T$ . The measure of this spanned lattice, normalized by  $\mu(T) = 1$ , is  $q^{-N \sum_j m_j}$ .*

*Proof.* Integral row operations send a primitive vector to any chosen standard basis vector. Choosing these operations independently in each factor gives every standard tensor basis vector multiplied by  $\pi^{\sum_j m_j}$ . Conversely all orbit vectors lie in this scaled lattice. Its index is  $q^{N \sum_j m_j}$ , giving the measure.  $\square$

**Proposition 6.4** (A strict simultaneous-action counterexample). *Over any commutative ring  $A$ , identify  $A^2 \otimes_A A^2$  with  $\text{Mat}_2(A)$ . The  $A$ -span of  $\{v \otimes v : v \in A^2\}$  is exactly the submodule of symmetric matrices. If  $A \neq 0$ , it omits  $E_{12}$ . For  $A = \mathcal{O}$  it has Haar measure zero in  $\text{Mat}_2(K)$ .*

*Proof.* Every generator has equal off-diagonal entries, a condition preserved by linear combinations. Conversely  $e_1 \otimes e_1, e_2 \otimes e_2$ , and

$$(e_1 + e_2) \otimes (e_1 + e_2) - e_1 \otimes e_1 - e_2 \otimes e_2 = E_{12} + E_{21}$$

span all symmetric matrices. No division by two occurs. The functional  $M \mapsto M_{12} - M_{21}$  vanishes on the span and takes value one on  $E_{12}$ . In the local-field case, the congruence  $M_{12} = M_{21} \pmod{\pi^t}$  occupies a proportion  $q^{-t}$  of the integral matrix lattice for every  $t$ . Its exact equality locus therefore has measure zero.  $\square$

The generating set in Proposition 6.4 is stable under every simultaneous action  $g \otimes g$ . Thus even the full simultaneous integral linear group does not provide Proposition 6.3. This rejects a specific abstract substitute for independent generation; it does not identify the actual IUT outputs with this diagonal example.

**6.2. A tensor point inside the holomorphic hull.** An important distinction changes the reachability question. The hull in [12, Section 9.10.7, p. 125] is, after the finite-étale decomposition of a tensor algebra, a product of principal local ideals. It is therefore a module over the *whole product of integer rings*. Independent linear actions are not necessary for the following containment.

Let  $E_1, \dots, E_m$  be finite extensions of  $\mathbb{Q}_p$ , and write

$$T = \bigotimes_{i=1}^m E_i = \prod_{\alpha} F_{\alpha}, \quad B = \prod_{\alpha} \mathcal{O}_{F_{\alpha}}, \quad A = \bigotimes_{i=1}^m \mathcal{O}_{E_i} \subset T,$$

where the integral tensor product is over  $\mathbb{Z}_p$ . Tensoring integral bases embeds  $A$  as a full lattice. Its elements are integral over  $\mathbb{Z}_p$ , so  $A \subset B$ , of finite index  $I = [B : A]$ .

**Proposition 6.5.** *For nonzero  $\tau_i \in E_i$ , put  $t = \bigotimes_i \tau_i$ . If a  $B$ -submodule  $H \subset T$  contains a set  $S$  with  $t \in S$ , then*

$$\bigotimes_i \tau_i \mathcal{O}_{E_i} = tA \subset tB \subset H.$$

*Both displayed lattices have full rank, and the smallest  $B$ -module containing the singleton  $\{t\}$  is exactly  $tB$ .*

*Proof.* The inverse of  $t$  is  $\bigotimes_i \tau_i^{-1}$ , by multiplication in the tensor algebra. Closure under  $B$  gives  $tB \subset H$ . The inclusion  $A \subset B$  and the tensor multiplication rule give the other equality and inclusion. Multiplication by the unit  $t$  is a  $\mathbb{Q}_p$ -linear automorphism and preserves full rank. Finally  $tB$  itself is a  $B$ -module containing  $t$ , proving the singleton assertion.  $\square$

The conclusion applies *after* taking the specified holomorphic hull. It does not put  $tA$  in the raw tensor-image set or its ordinary  $\mathbb{Z}_p$  convex closure. As printed, Section 9.10.7 requires the component generators to be nonzero and integral. An arbitrary bounded nonintegral set is not covered by that definition. With fractional generators allowed, a bounded set containing a unit has a unique such hull: take the least valuation in each nonzero component. This is an explicit extension of the convention, not an assumption that every source output is integral.

**Proposition 6.6** (The two closures can differ). *Let  $E = \mathbb{Q}_3(\pi)$  with  $\pi^2 = 3$ . Under*

$$E \otimes_{\mathbb{Q}_3} E \simeq E^2, \quad a \otimes b \longmapsto (ab, \sigma(a)b), \quad \sigma(\pi) = -\pi,$$

*the tensor order satisfies*

$$A = \{(x, y) \in \mathcal{O}_E^2 : x - y \in \pi \mathcal{O}_E\}, \quad [\mathcal{O}_E^2 : A] = 3.$$

*It is already a closed  $\mathbb{Z}_3$ -module, whereas its holomorphic hull is  $B = \mathcal{O}_E^2$ . The singleton  $(1, 1)$  also has holomorphic hull  $B$ , while its ordinary convex closure is contained in  $\mathbb{Z}_3(1, 1)$ .*

*Proof.* Valuations show that  $\pi \notin \mathbb{Q}_3$  and  $\mathcal{O}_E = \mathbb{Z}_3 + \mathbb{Z}_3\pi$ . The images of an integral tensor basis are

$$(1, 1), \quad (\pi, -\pi), \quad (\pi, \pi), \quad (3, -3).$$

Their differences are divisible by  $\pi$ . Conversely write  $x = a + b\pi$ ,  $y = c + d\pi$ . The asserted congruence is  $3 \mid a - c$ , and coefficients  $(a+c)/2, (b-d)/2, (b+d)/2, (a-c)/6$  express  $(x, y)$  in the displayed basis. Reduction of  $x - y$  modulo  $\pi$  gives the quotient  $\mathbf{F}_3$  and hence the index. Both  $A$  and the singleton contain  $(1, 1)$ ; its  $B$ -multiples are all of  $B$ . The compact module  $\mathbb{Z}_3(1, 1)$  is convex and omits  $(1, 0)$ , proving the last distinction without relying on a choice between affine and absolutely convex conventions.  $\square$

This gives a counterexample to the general equality of holomorphic hull and minimum ordinary convex closure in [12, Proposition 9.10.8.1]. It does not refute IUT or abc. In particular the positive containment in Proposition 6.5 must be used on the holomorphic-hull side of this distinction.

**6.3. Native coefficients and compatible volumes.** There is a coherent marking-preserving collation of the relevant local Kummer classes. For a fixed finite extension  $E/\mathbb{Q}_p$  and marked copies  $i_y : E \simeq E_y$ , Kummer theory identifies the integral multiplicative cohomology with

$$\mathcal{K}(E_y) = \varprojlim_n E_y^\times / (E_y^\times)^{p^n}.$$

This is the identification used in [11, Proposition 7.2.2]. At each finite level, the map  $i_z i_y^{-1}$  induces an isomorphism of these quotients; their inverse limit gives a topological isomorphism  $F_{y \rightarrow z}$  with

$$F_{y \rightarrow z}(\kappa_y(i_y(u))) = \kappa_z(i_z(u)), \quad F_{z \rightarrow r} F_{y \rightarrow z} = F_{y \rightarrow r}.$$

It preserves the unit-completion subgroup and can be chosen coherently every time a label is repeated. This uses one family of the factorwise topological collations in the cited source; it does not assert that every such collation preserves a marked arithmetic element.

For  $b_p = p$  when  $p$  is odd and  $b_2 = 4$ , the coefficient in [12, equation (9.7.2.2)] is

$$\lambda_{b_p}(a) = \frac{\log(1 + b_p a)}{b_p}, \quad a \in \mathcal{O}_E.$$

For  $a \neq 0$  it is nonzero and has the same valuation as  $a$ . Indeed, with  $v(p) = 1$  and  $x = b_p a$ , every term after the first in the logarithm series has greater valuation:

$$v(x^n/n) - v(x) = (n-1)v(x) - v_p(n) > 0 \quad (n \geq 2).$$

For odd  $p$  use  $v_p(n) < n-1$ , and for  $p = 2$  use  $v(x) \geq 2$ . The terms tend to zero, so  $v(\log(1+x)) = v(x)$ . The field markings commute with this convergent series, yielding the same normalized coefficient in the fixed target. In particular, the background coefficient  $\lambda_{b_p}(1)$  is a unit.

**Proposition 6.7.** Put  $d_i = [E_i : \mathbb{Q}_p]$ ,  $D_T = \prod_i d_i$ , and normalize Haar measure on  $T$  by  $\mu(A) = 1$ . For  $t = \bigotimes_i \tau_i$ ,

$$\mu(tB) = I \prod_i |\tau_i|_{E_i}^{D_T/d_i}, \quad |x|_{E_i} = |N_{E_i/\mathbb{Q}_p}(x)|_p.$$

For any  $c_0 > 0$ , the functional  $\nu(U) = \mu(U)^{c_0/D_T}$  is monotone, and every measurable  $H \supset tB$  satisfies

$$\nu(H) \geq I^{c_0/D_T} \prod_i |\tau_i|_{E_i}^{c_0/d_i} \geq \prod_i |\tau_i|_{E_i}^{c_0/d_i}.$$

A factor prescription  $\prod_i \mu_i(V_i)^{\gamma_i}$  for the same integral tensor lattice can be independent of its presentation only if all  $d_i \gamma_i$  agree. These equalities are also sufficient on full tensor lattices.

*Proof.* The  $I$  translates of  $A$  partition  $B$ , so  $\mu(B) = I$ . Multiplication by  $\tau_i$  in its tensor factor has determinant  $N_{E_i/\mathbb{Q}_p}(\tau_i)^{D_T/d_i}$ . Multiplication by  $t$  is the composition of these maps, giving the displayed Haar factor. Monotonicity and the positive power give the inequalities. For necessity of balancing, put a scalar  $p$  in any one factor of the tensor lattice  $pA$ . The proposed values are  $p^{-d_i\gamma_i}$  and must agree. If  $d_i\gamma_i = c_0$ , the normalized functional  $\mu^{c_0/D_T}$  gives the factor prescription by the same determinant computation, proving sufficiency.  $\square$

The normalization matters when using an upper estimate from another source. Denote the preceding Haar measure by  $\mu_A$ , and instead let  $\mu_B(B) = 1$ . On a measurable region of finite positive measure, put

$$V_A(U) = \frac{\log \mu_A(U)}{D_T}, \quad V_B(U) = \frac{\log \mu_B(U)}{D_T}.$$

Uniqueness of Haar measure and  $\mu_A(B) = I$  give

$$(26) \quad V_A(U) = V_B(U) + \frac{\log I}{D_T}, \quad V_B(tB) = \sum_i \frac{\log |\tau_i|_{E_i}}{d_i}.$$

The positive power of Haar measure used above need not itself be an additive measure. Only its monotonicity was asserted.

This conversion is required by the original texts: [12, Section 9.10.3.1] assigns value one to the tensor order  $A$ , whereas [19, Proposition 1.4(i), author-version p. 13] normalizes the log-volume of the maximal order  $B$  to zero. The positive index contribution cannot be carried into a  $B$ -normalized upper estimate. The second equality in (26) preserves the native lower bound without that contribution.

For example, if all  $m$  factors equal a Galois extension  $E/\mathbb{Q}_p$  of degree  $d$ , then

$$E^{\otimes m} \simeq E^{d^{m-1}}, \quad \log_p[B : A] = \frac{(m-1)d^{m-1}\delta_E}{2},$$

where  $\delta_E$  is the exponent of its discriminant over  $\mathbb{Z}_p$ . Indeed, the trace Gram matrix on a tensor basis has discriminant exponent  $md^{m-1}\delta_E$ . On the product maximal order that exponent is  $d^{m-1}\delta_E$ . Their difference is twice the exponent of the lattice index. Thus

$$V_A(U) - V_B(U) = \frac{(m-1)\delta_E}{2d} \log p.$$

For copies of a fixed selected field  $E = L'_w$  over  $F = L_{\text{mod},v}$ , the source weight  $\gamma = 1/[E : F]$  obeys  $[E : \mathbb{Q}_p]\gamma = [F : \mathbb{Q}_p]$ . It is thus balanced. A tuple with  $r$  theta coefficients  $\lambda_{b_p}(a)$  and the remaining coefficients  $\lambda_{b_p}(1)$  gives the native lower bound

$$\nu(H) \geq |a|_E^{r/[E:F]} \geq |a|_E^r \quad \text{if } 0 < |a|_E < 1.$$

For a rational Frey curve  $C$ , the field of moduli is  $\mathbb{Q}$ . The underlying punctured curve is defined over  $\mathbb{Q}$ ; its absolute field of moduli, used before adjoining level structure in [12, Section 3.1(4)], is therefore  $\mathbb{Q}$ . Equivalently the Legendre parameter  $-b/a$  gives  $\mathbb{Q}(j) = \mathbb{Q}$  in [13, Section 5.3]. The later extensions  $L = \mathbb{Q}(i, C[30])$  and  $L' = L(C[\ell])$  are Galois over  $\mathbb{Q}$  and do not redefine that unlevelled moduli field. Choosing one place of  $L'$  above each rational prime gives the place-compatible selected bijection of [12, Section 3.3(14) and Proposition 9.4.2.4(4)]. Thus there is one selected place per prime. The internal field decomposition of  $E^{\otimes m}$  still remains.

For a split multiplicative place with  $q \in \mathbb{Q}_p$ , take a root  $a^{2\ell} = q$  in the selected field  $E$ . The native module absolute value satisfies  $|q|_E = |q|_p^d$ , so

$$|\lambda_{b_p}(a)|_E^{1/d} = |q|_p^{1/(2\ell)}.$$

A block with one such entry and background units therefore gives

$$(27) \quad V_B(H) \geq V_B(tB) = -Q_p, \quad Q_p = -\frac{\log |q|_p}{2\ell} > 0.$$

The chosen marking-induced isomorphisms give an actual tuple in the collation, and convex closure retains it. The tuple-to-tensor map and the module hull then give the stated containment. No independent action on repeated labels is needed.

**6.4. Changing the target of the whole collation.** Fix a family of source groups  $C_y$ , a set of configurations of source classes, and a fixed slot set in each block. Require repeated occurrences of a source label to use the same isomorphism. Let  $\Psi_k$  be the union of all resulting tuples in a target  $C_k$ , where all allowed topological group isomorphisms are used. The same discussion applies to a groupoid of Galois-induced isomorphisms.

**Proposition 6.8** (Covariance for a fixed source family). *If  $c : C_0 \simeq C_1$  is an allowed target isomorphism, then*

$$\Psi_1 = c^{\text{slots}}(\Psi_0).$$

*If  $c$  in logarithmic coordinates is induced by a local field marking, the same covariance holds for closed convex closure, tensor image, and the maximal-order module hull. In particular the transported hulls have equal  $V_A$  volumes, and equal  $V_B$  volumes with the respective normalizations.*

*Proof.* Postcomposition  $f \mapsto c \circ f$  is a bijection from  $\text{Iso}(C_y, C_0)$  to  $\text{Iso}(C_y, C_1)$ , with inverse postcomposition by  $c^{-1}$ . Apply it to the whole family of maps used by a source configuration. Equal maps on repeated labels remain equal in both directions. Taking the union proves the set equality. For a groupoid the same argument uses closure under composition and inverse.

The marking induces a linear homeomorphism in logarithmic coordinates, so it carries a smallest closed convex region to the corresponding smallest closed convex region. Tensoring the field maps commutes with the map on pure tensors. The resulting algebra isomorphism preserves integrality over  $\mathbb{Z}_p$  and thus carries  $A_0, B_0$  to  $A_1, B_1$ . Minimality of the coefficient-module hull, applied also to the inverse, proves equality of the transported hulls. Finally uniqueness of Haar measure transports the chosen normalization, and dimensions are unchanged.  $\square$

The source family and its classes are hypotheses of this proposition. It justifies a change of target in the all-isomorphism construction of [12, Section 9.7.5.1]; it does not show that every operation intended by a later Frobenius comparison keeps that source family fixed. In particular a favorable native point and covariance for this family cannot be combined with an upper estimate for a different family.

To see the normalization issue directly, write a rescaled untilt norm on the marked finite field as  $N_y(x) = |x|_E^{s_y}$ ,  $s_y > 0$ . The native Haar contribution is

$$|\lambda_{b_p}(a)|_E^{1/d} = N_y(\lambda_{b_p}(a))^{1/(ds_y)}.$$

Changing  $s_y$  alone leaves it unchanged. Also replacing  $q$  by  $q^p$  rebuilds the principal unit from  $a^p$  and gives coefficient  $\lambda_{b_p}(a^p)$ , whereas raising  $1 + b_p a$  to its  $p$ th power gives  $p\lambda_{b_p}(a)$ . They need not agree: for  $p = 3, E = \mathbb{Q}_3, a = 3$ , their valuations are three and two, respectively. The Frobenius in [11, Corollary 5.6.2] acts on a perfected multiplicative monoid, not by definition on this fixed integral Kummer lattice.

The original IUT IV computation explicitly uses a squared exponent: [19, Theorem 1.10, Step (v), author-version pp. 27–29] takes the  $j$ th theta-pilot valuation to be  $j^2 v_p(q)/(2\ell)$ . Replacing the coefficient above by  $\lambda_{b_p}(a^{j^2})$  would give the single-point lower bound  $-j^2 Q_p$ , rather than  $-Q_p$ . For  $s = (\ell - 1)/2$  the average of these coefficients is

$$\frac{1}{s} \sum_{j=1}^s j^2 = \frac{(s+1)(2s+1)}{6} = \frac{\ell(\ell+1)}{12}.$$

Proposition 6.8 can be proved for either fixed family, but it does not identify them. The required favorable lower bound for the correctly powered possible-image family is a further assertion.



**6.5. The source weights for several selected places.** The mixed-place weights can also be read from the original upper source, [19, Remark 1.7.1, author-version p. 17]. Let  $F = L_{\text{mod}}$  have degree  $n$ , put  $h_v = [F_v : \mathbb{Q}_p]$ ,  $d_v = [L'_{w(v)} : \mathbb{Q}_p]$ , and  $\gamma_v = h_v/d_v$  for  $v \mid p$ . Then  $\sum_v h_v = n$ . For a word  $f$  of length  $m$  in these places put

$$D_f = \prod_i d_{f(i)}, \quad \rho_f = \frac{\prod_i h_{f(i)}}{n^m}.$$

With  $B_f$  the maximal order in the corresponding tensor algebra, the degree-normalized functional on a product of ideal hulls is

$$V_{B,p,m}(H) = \sum_f \frac{\rho_f}{D_f} \log \mu_{B_f}(H_f), \quad \mu_{B_f}(B_f) = 1.$$

The word weights sum to one and have marginal  $h_v/n$  at every slot. Applying the tensor determinant identity therefore gives

$$V_{B,p,m}(tB) = \frac{1}{n} \sum_i \sum_{v \mid p} \gamma_v \log |\tau_{i,v}|_{L'_{w(v)}}.$$

The corresponding tensor-order normalization differs by exactly  $\sum_f (\rho_f/D_f) \log[B_f : A_f]$ . This proves the formula on product ideal hulls. It is not a product formula for an arbitrary measurable non-product region. The extra global factor  $n$  must be coordinated with the arithmetic-degree convention; it equals one in the rational branch.

These constructions prove the local point-to-lattice implication and fixed-source target covariance with their stated normalizations. They do not yet identify the full Frobenius-shifted source families in [13, Theorem 6.10.1] with those of the later upper estimate. That estimate must concern the same hull and volume functional. Part III already introduces the volume of a containing compact region on p. 121; omission of a hull symbol in a later formula is therefore not, by itself, evidence that a different set was estimated. What is needed is the actual comparison with the region and measure in IUT IV, rather than an identification by notation.

**6.6. What a source-derived application must establish.** The July 2026 LANA report [15] isolates an identification between q-pilot and possible-output volume data in the passage to IUT III, Corollary 3.12 [16]. Joshi's July 30 response [14] refers to distinct arithmeticoids and his Part IV comparison [13, Theorem 6.10.1]. These are source claims to reconstruct, not assumptions used in our local theorems.

Part III [12, Proposition 9.7.5.1] allows specified topological isomorphisms in its collation. Section 6.2 constructs one marking-preserving family and proves containment after the stated holomorphic hull. It does not require unrestricted independent linear actions. The remaining identification must compare the same set of source objects, the same target region, and the same numerical normalization with the later upper estimate.

For a source-faithful finite packet of local sets  $\mathcal{R}_v \subset \mathcal{O}_v^{N_v}$  and positive weights  $\lambda_v$ , Theorem 6.1 gives, when all defects are finite,

$$V_H = - \sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v.$$

Suppose genuine local target regions  $\Theta_v$  contain the respective hulls and have finite positive measure. Define, on these same carriers,  $V_\Theta = \sum_v (\lambda_v/N_v) \log \mu_v(\Theta_v)$ . Let  $V_q \in \mathbb{R}$  be the fixed native q-pilot log-volume under the same source normalization. Then

$$\sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v \leq -V_q \implies V_q \leq V_H \leq V_\Theta.$$

The final inequality follows from local measure monotonicity, followed by the logarithm and the positive weighted sum. A different volume functional on a non-product region requires a separate monotonicity theorem. This is a sufficient numerical interface. The native points and their holomorphic hulls above provide one local construction; identifying them with the region

bounded in IUT IV [19], controlling the same normalized determinant sum, and deriving the uniform global comparison remain separate obligations. Archimedean factors and infinite-packet convergence are not covered by the finite local theorem.

## 7. ADMISSIBLE UNIT ACTIONS AND A CYCLOTOMIC TRACE INVARIANT

This section uses the actual all-open-subgroup condition in [18, Example 1.8(iv), author-version pp. 38–39]. It must be distinguished from allowing every automorphism of one fixed integral lattice. Fix a finite extension  $E/\mathbb{Q}_p$ , an algebraic closure  $\overline{E}$ , and  $G = \text{Gal}(\overline{E}/E)$ . For each finite  $K/E$  inside  $\overline{E}$ , put

$$U_K = \mathcal{O}_K^\times, \quad L_K = \log U_K \subset K, \quad I_K = p^{-1}L_K.$$

The logarithm is used only on units. Its kernel on  $U_K$  is the finite group of roots of unity in  $K$ , and  $L_K$  is a full  $\mathbb{Z}_p$ -lattice in  $K$ . These assertions hold also for  $p = 2$ .

Logarithm induces a  $G$ -equivariant additive isomorphism

$$(28) \quad \mathcal{O}_{\overline{E}}^\times / \mu(\overline{E}) \simeq (\overline{E}, +).$$

Indeed the finite-field kernel statement proves injectivity. Given  $x \in \overline{E}$ , choose  $n$  large enough for  $\exp(p^n x)$  to converge in a finite extension and choose a  $p^n$ -th root  $y$  of it. Then  $y$  is a unit and  $\log y = x$ , proving surjectivity. If  $H = \text{Gal}(\overline{E}/K)$ , the invariants of the quotient in (28) are  $K$ , whereas the image of the invariants of the original unit group is precisely  $L_K$ .

Under this identification, the group  $\text{Ism}(G)$  specified in [18] consists of additive  $G$ -equivariant automorphisms  $F$  of  $\overline{E}$  preserving  $L_K$  for *every* finite  $K/E$ . The source requires preservation of these lattices, not only of their rational spans. Such a map is  $\mathbb{Q}_p$ -linear on each  $K$ : preservation of  $p^n L_K$  implies continuity, which extends integer linearity to  $\mathbb{Z}_p$ -linearity and then to  $\mathbb{Q}_p$ -linearity.

### 7.1. The all-open condition forces scalar action.

**Lemma 7.1** (Realizing logarithmic sublattices by norms). *For every finite-index  $\mathbb{Z}_p$ -submodule  $J \subset L_E$ , there is a finite abelian extension  $K/E$  such that*

$$\text{Tr}_{K/E}(L_K) = J.$$

*Proof.* Let  $V = \log^{-1}(J) \subset U_E$ , choose a uniformizer  $\pi$  of  $E$ , and put  $N = \pi^{\mathbb{Z}}V$ . This is an open finite-index subgroup of  $E^\times$ . The local existence theorem of class field theory [21, Chapter I, Theorem 1.4] supplies a finite abelian  $K/E$  with  $\text{Norm}_{K/E}(K^\times) = N$ . For integer-normalized valuations,

$$v_E(\text{Norm}_{K/E} z) = f_{K/E} v_K(z).$$

Consequently a norm is a unit exactly when its preimage is a unit, and  $\text{Norm}_{K/E}(U_K) = N \cap U_E = V$ . The identity  $\log \text{Norm} z = \text{Tr} \log z$  follows by summing the logarithms of the conjugates. Therefore  $\text{Tr}_{K/E}(L_K) = \log V = J$ .  $\square$

**Lemma 7.2** (Sublattice rigidity). *An automorphism of a nonzero finite free  $\mathbb{Z}_p$ -module preserving every finite-index submodule is multiplication by one element of  $\mathbb{Z}_p^\times$ .*

*Proof.* Choose a basis  $e_1, \dots, e_d$ . Preservation of  $\mathbb{Z}_p e_i + p^n L$  for every  $n$  puts the image of  $e_i$  in their intersection, which is  $\mathbb{Z}_p e_i$ : every other coordinate is divisible by every power of  $p$  and hence is zero. Write  $F(e_i) = \lambda_i e_i$ . The same argument for  $e_i + e_j$  implies  $\lambda_i = \lambda_j$ . Thus  $F = \lambda \text{id}$ ; surjectivity shows that  $\lambda$  is a unit. The rank-one case is already covered by the first step.  $\square$

**Theorem 7.3** (Classification of the specified Ism action). *For the group defined in [18, Example 1.8(iv)],*

$$\text{Ism}(G) = \mathbb{Z}_p^\times$$

*under (28), with the right side acting by scalars.*

*Proof.* For finite abelian  $K/E$ , equivariance gives  $F_E \operatorname{Tr}_{K/E} = \operatorname{Tr}_{K/E} F_K$ . Since  $F_K(L_K) = L_K$ , Lemma 7.1 implies that  $F_E$  preserves every finite-index submodule of  $L_E$ . Lemma 7.2 gives  $F_E = \lambda_E \operatorname{id}$  with  $\lambda_E \in \mathbb{Z}_p^\times$ . Repeat over any finite  $K/E$ : the hypothesis includes all finite extensions of  $K$ , since their Galois groups are open in  $G$ . Thus  $F_K = \lambda_K \operatorname{id}$ . Restriction to the nonzero subfield  $E$  gives  $\lambda_K = \lambda_E$ , proving the assertion on all of  $\overline{E}$ . Conversely every  $\mathbb{Z}_p$ -unit scalar preserves all the lattices and commutes with  $G$ . It is the image of a unit of  $\widehat{\mathbb{Z}}$  under the unit-power action; prime-to- $p$  components act trivially after quotienting by all roots of unity.  $\square$

Only preserving  $L_E$  would leave the full group  $\operatorname{GL}_{[E:\mathbb{Q}_p]}(\mathbb{Z}_p)$ ; the finite-extension hypothesis is decisive. The full-unit lifting theorem [9, Theorem 7.6] by itself does not classify automorphisms of the torsion quotient. The preceding proof supplies that additional step. For the indeterminacy called Ind2 in [16, Theorem 3.11(i)], the theorem applies to its indicated Ism factors. It does not classify the Galois arrows of Ind1 or the set operations of Ind3.

For example, if  $1 < e(E/\mathbb{Q}_p) \leq p-2$ , logarithm and exponential on the maximal ideal give  $I_E = \pi^{1-e} \mathcal{O}_E$ . The unit  $u = \log(1+p)/p$  has native valuation zero, and all its Ism images still have valuation zero. None reaches the least valuation  $1/e - 1$  of this larger lattice. This statement concerns Ism; it adds no unproved restriction to Ind1.

**7.2. What the procession retains, and an exact Ind2 hull equality.** The Ind1 source in [16, Theorem 3.11(i)] is the mono-analytic procession of [17, Proposition 6.9(ii)]. At a nonarchimedean place its retained category is  $\mathcal{B}(E)^0$ , the connected finite etale covers of  $E$ , rather than the earlier curve-covering category [17, Examples 3.2(i), 3.3(i), Definition 4.1(iii)–(iv)]. Its self-equivalences, up to natural isomorphism, correspond to  $\operatorname{Out}(G_E)$ : pullback of finite transitive  $G_E$ -sets constructs one direction, and adjoining finite disjoint unions and recovering the fibre functor constructs the inverse.

For fixed capsule labels, the capsule-full poly-isomorphism contains all collections of constituent isomorphisms [17, §0, pp. 33–34; Definition 4.10]. Consequently every local outer Galois automorphism occurs as a representative in an allowed poly-automorphism, with the same representative used at repeated labels if required. To see this directly, identify each labelled category with  $\mathcal{B}(E)^0$ , conjugate the chosen pullback equivalence to each copy, and use identity maps at the other places. Keep the capsule and stage injections equal to the identity. Each resulting collection belongs to the required full set of constituent isomorphisms, and conjugation preserves the full connecting sets. There is no further requirement to lift back to the curve category. This is a statement about representatives inside poly-morphisms, not a claim that each representative is a distinct object or morphism of the coarse category. It supplies no arbitrary integral linear lift.

**Proposition 7.4** (Ind2 does not enlarge a fixed module hull). *On a fixed marked nonarchimedean tensor carrier, let*

$$T = \bigotimes_{i, \mathbb{Q}_p} \left( \prod_{v \in J_i} E_{i,v} \right), \quad A = \bigotimes_{i, \mathbb{Z}_p} \left( \prod_{v \in J_i} \mathcal{O}_{E_{i,v}} \right) \subset B,$$

where  $B$  is the maximal order of the finite etale algebra  $T$ . For any subset  $S \subset T$ , write  $\Gamma_2 S$  for the union of its images under the local Ind2 operators of [16, Theorem 3.11(i)]. Then

$$\operatorname{span}_B(\Gamma_2 S) = \operatorname{span}_B(S).$$

The same equality holds for closed  $B$ -module hulls.

*Proof.* Theorem 7.3 makes the action on each local direct summand multiplication by a scalar  $\lambda_{i,v} \in \mathbb{Z}_p^\times$ . In factor  $i$ , their tuple  $c_i = (\lambda_{i,v})_v$  is a unit of  $\prod_v \mathcal{O}_{E_{i,v}}$ . The tensor product operator is multiplication by  $c = \bigotimes_i c_i \in A^\times$ , with inverse the tensor of the  $c_i^{-1}$ . Thus every image of  $S$  lies in  $\operatorname{span}_B(S)$ . The identity Ind2 operator supplies the reverse inclusion of spans. Taking closures gives the closed version. Any extra convention forcing some scalars to coincide leaves the argument unchanged.  $\square$

This removes the Ind2 step from a  $B$ -module-hull computation once the Ind1 image set and native markings are fixed. It does not remove the distinction between  $A$  and  $B$ , turn an Ind1 map into a  $B$ -linear map, or identify Ind3 with multiplication by a unit.

**7.3. Integral Galois transport preserves trace depth.** A coefficient-compatible Galois arrow consists of a continuous isomorphism  $\alpha : G_{E'} \simeq G_E$  and an isomorphism  $\beta : \alpha^* \mathbb{Z}_p(1)_E \simeq \mathbb{Z}_p(1)_{E'}$  of integral Tate modules. The cyclotomic character and inertia are group-theoretically determined [22, Propositions 1.1–1.2 and Corollary 1.3]. In particular the coefficient isomorphism can be chosen integrally; one cannot insert multiplication by  $p$  as an isomorphism.

**Proposition 7.5.** *The induced logarithmic map  $F_{\log} : E \rightarrow E'$  satisfies*

$$(29) \quad \mathrm{Tr}_{E'/\mathbb{Q}_p}(F_{\log}x) = c_\alpha \mathrm{Tr}_{E/\mathbb{Q}_p}(x) \quad (x \in E)$$

for some  $c_\alpha \in \mathbb{Z}_p^\times$ . The statement also holds with both logarithmic coordinates divided by  $p$ .

*Proof.* Put  $h_E = \log_p \chi_E \in H^1(G_E, \mathbb{Z}_p)$ , where the latter coefficient action is trivial. Equivariance of  $\beta$  gives  $\chi_E \alpha = \chi_{E'}$ , hence  $\alpha^* h_E = h_{E'}$ . Local duality and the invariant map identify  $H^2(G_E, \mathbb{Z}_p(1))$  with  $\mathbb{Z}_p$ . Integral cohomology transport is an isomorphism, so it acts there by a unit  $c_\alpha$ , not by an arbitrary nonzero rational  $p$ -adic scalar.

The local reciprocity cup-product formula, for an integral character  $h$  and a Kummer class  $\kappa(z)$ , is

$$\mathrm{inv}(h \cup \kappa(z)) = h(\mathrm{rec}_E z).$$

It follows at finite level from the cyclic-algebra description and then by inverse limit; see [21, Chapter III, Proposition 3.6 and §4]. In the arithmetic-Frobenius convention,

$$h_E(\mathrm{rec}_E z) = -\log_p \mathrm{Norm}_{E/\mathbb{Q}_p} z = -\mathrm{Tr}_{E/\mathbb{Q}_p} \log_p z \quad (z \in U_E).$$

The integral unit Kummer submodule is also preserved: it is the annihilator, under this pairing, of the unramified  $\mathbb{Z}_p$ -character. Characteristic inertia carries a generator of that unramified character module to a unit multiple of a generator, and cup-product naturality preserves its annihilator.

Applying the same naturality with  $h_E$  now proves (29) on  $L_E$ . Torsion unit classes are killed by logarithm and pair to zero in the torsion-free integral  $H^2$ ; the argument therefore descends to the actual free unit lattice. Since  $L_E$  spans  $E$ ,  $\mathbb{Q}_p$ -linearity proves the formula everywhere. Uniform division by  $p$  on both sides does not change it.  $\square$

Composing with the Ism factors of Theorem 7.3 only changes  $c_\alpha$  by a unit. Thus the valuation of a nonzero absolute trace is an invariant of these local Galois/Kummer/Ism arrows. For example, on  $E = \mathbb{Q}_7(\pi)$  with  $\pi^3 = 7$ , the normalized integral log lattice  $I_E$  has basis  $1, \pi/7, \pi^2/7$ . Swapping its first two basis vectors is not an admissible Galois/Kummer arrow: their traces are 3 and 0. No claim that every matrix satisfying (29) is admissible is made.

**Theorem 7.6** (A native root-power obstruction). *Let  $p \neq \ell$  be odd primes,  $b \in \mathbb{Q}_p$  with  $N = v_p(b) > 0$  and  $\ell \nmid N$ , and let  $E/\mathbb{Q}_p$  contain  $\mu_\ell$  and  $a$  with  $a^\ell = b$ . Write  $d = [E : \mathbb{Q}_p]$  and, for  $s \geq 1$  with  $\ell \nmid s$ , put  $\tau_s = \log(1 + pa^s)/p$ . Then*

$$(30) \quad \mathrm{Tr}_{E/\mathbb{Q}_p} \tau_s = \frac{d}{\ell p} \log(1 + p^\ell b^s),$$

$$(31) \quad v_p(\mathrm{Tr}_{E/\mathbb{Q}_p} \tau_s) = v_p(d/\ell) + \ell - 1 + sN.$$

*In particular, distinct such  $s$  give distinct orbits under the arrows of Proposition 7.5, even after Ism action.*

*Proof.* The field  $F = \mathbb{Q}_p(\mu_\ell)$  is unramified over  $\mathbb{Q}_p$ . The valuation assumption makes  $b$  a non- $\ell$ -th-power in  $F$ , so  $K = F(a)$  has degree  $\ell$  over  $F$ . Since multiplication by  $s$  permutes the exponents modulo  $\ell$ , and  $\ell$  is odd,

$$\mathrm{Norm}_{K/F}(1 + pa^s) = \prod_{\zeta \in \mu_\ell} (1 + p\zeta a^s) = 1 + p^\ell b^s.$$

All factors are principal units. Taking logarithms, dividing by  $p$ , and using trace transitivity gives (30): the remaining factor is  $[E : K][F : \mathbb{Q}_p] = d/\ell$ . For  $x = p^\ell b^s$ , the term  $x$  is the unique least-valuation term in  $\log(1 + x)$ , since

$$v_p(x^k/k) - v_p(x) = (k-1)(\ell + sN) - v_p(k) > 0 \quad (k \geq 2).$$

This proves (31), including nonvanishing. Its values differ by  $(s-t)N$  for distinct labels, which (29) cannot change.  $\square$

For a concrete example take  $p = 11$ ,  $\ell = 5$ ,  $b = 11$ , and  $E = \mathbb{Q}_{11}(\pi)$  with  $\pi^5 = 11$ . The base field contains  $\mu_5$ ; the trace depths of  $\tau_1$  and  $\tau_4$  are 5 and 8. This is a specified local example, not a construction of all the global initial theta data.

There is a direct local link with a split Tate curve having rational full 2-torsion. If  $q$  is its parameter, choose  $b^2 = q$ . The class of  $b$  in  $\overline{\mathbb{Q}_p}^\times/q^\mathbb{Z}$  is rational, so  $g(b)/b \in q^\mathbb{Z} \cap \{1, -1\}$ . Valuation makes this intersection  $\{1\}$ ; hence  $b \in \mathbb{Q}_p$ . Rationalizing full  $\ell$ -torsion also rationalizes full  $2\ell$ -torsion and supplies the required  $a$  and  $\mu_\ell$ . The theorem consequently applies when  $\ell \nmid v_p(q)/2$ . It does not assert that every bad prime meets that condition.

For a pure tensor with  $m-1$  factors  $u = \log(1+p)/p$  and one factor  $\tau_s$ , trace in  $E^{\otimes m}$  is the product of the factor traces. Its valuation is

$$(m-1)v_p(d) + v_p(d/\ell) + \ell - 1 + sN.$$

Local arrows, Ism actions, and factor permutations preserve this number, including when labels require repeated use of one arrow. This separates the exponent-1 input from the exponent- $j^2$  input for  $j > 1$ ,  $j < \ell$ . Raising  $1+pa$  to the power  $j^2$ , which scales its logarithm, is a different operation from replacing  $a$  by  $a^{j^2}$ .

**7.4. Why this is not yet a holomorphic-hull comparison.** Multiplying generators by coefficients of the maximal tensor order need not preserve their trace depths. Thus distinct pure orbits do not imply distinct module hulls. The upper-semicompatibility operation Ind3 in [16, Theorem 3.11(ii)] is a further operation, and has not been replaced here by an automorphism of one fixed Kummer module. Nor does changing an untilt's real-valued norm change the native trace in a fixed marked  $p$ -adic field.

A converse test is useful: trace preservation alone permits a common least-layer displacement, without establishing an Ind1 lift.

**Lemma 7.7** (Common avoidance with fixed trace). *Let  $k$  be a field of characteristic different from two, let  $V$  be a  $k$ -vector space, and let  $t, l \in V^*$ . Suppose  $t(w) = 0$ ,  $l(w) = 1$ . For any nonzero  $x, y \in V$ , there is an automorphism  $F$  with  $tF = t$  and  $l(Fx), l(Fy) \neq 0$ .*

*Proof.* Set  $P(v) = v - l(v)w$ . Choose a functional  $h$  nonzero on each nonzero vector among  $P(x), P(y)$ . Such a functional exists: separate the first vector; if that functional kills the second, separate the second and use either the new functional or the sum, according to its value on the first. Put  $f = hP$ , so  $f(w) = 0$ . The maps  $F_z(v) = v + zf(v)w$  have inverses  $F_{-z}$  and preserve  $t$ . For  $v = x, y$ , the polynomial  $l(v) + zf(v)$  is nonzero: if  $l(v) = 0$ , then  $P(v) = v \neq 0$ . Two nonzero affine polynomials have at most two roots altogether. The three distinct scalars 0, 1,  $-1$  therefore contain a valid choice of  $z$ .  $\square$

For finite Galois  $E/\mathbb{Q}_p$  with  $2 \leq e \leq p-2$ , one has  $I_E = \pi^{1-e}\mathcal{O}_E$ , the inverse different. Both  $\text{Tr} : I_E \rightarrow \mathbb{Z}_p$  and  $\text{Tr} : \mathcal{O}_E \rightarrow \mathbb{Z}_p$  are onto; the latter follows by tracing first through the tame ramified extension and then through the unramified one. On  $I_E/pI_E$ , trace modulo  $p$  is independent of any nonzero  $\mathbb{F}_p$ -coordinate of  $I_E/\pi I_E$ : the latter vanishes on  $\mathcal{O}_E \subset \pi I_E$ , whereas trace does not. The lemma gives an automorphism preserving trace modulo  $p$  and reaching the least valuation layer for two primitive vectors simultaneously. Choose a basis splitting integral trace. The residue matrix has first row  $(1, 0, \dots, 0)$ ; lift that row exactly and the others arbitrarily. The determinant is a unit, so this is an integral invertible lift preserving trace exactly. No lift to an absolute Galois automorphism is established by this construction. Determining that image, and then performing all source-prescribed hull operations, remains

necessary for a global comparison. No IUT theorem or the abc conjecture has been disproved by these local nonreachability statements.

## 8. EFFECTIVE BOUNDS AND FINITENESS IN A SPECIFIED PELL FAMILY

We now restrict to positive integers  $b, A, B, u, v, r_0, s, X$  and a prime  $p \geq 37$  satisfying

$$(32) \quad b = Au^2, \quad b+1 = Bv^2, \quad b+2 = 3r_0^2, \quad b+3 = s^2,$$

$$(33) \quad D = 3AB, \quad Z = b^2 + 3b + 1 = T_p(X), \quad D+1 \leq X^2, \quad X^p \leq Z.$$

Here  $T_p$  is the Chebyshev polynomial of the first kind. These equations define the specified residual packet; no reduction of general abc counterexamples to it is asserted. Set

$$H = \log(b+2), \quad L = \log(D+1), \quad C_0 = 12^{331776}, \quad K_0 = \log(2C_0).$$

The packet immediately yields

$$(34) \quad (D+1)^p \leq X^{2p} \leq Z^2 < (b+2)^4, \quad pL < 4H.$$

Theorem 2.2 of Bérczes–Evertse–Győry [2], specialized to  $K = \mathbb{Q}$ ,  $S = \{\infty\}$ , and exponent two, gives the following published input. If  $f \in \mathbb{Z}[t]$  is squarefree of degree  $n \geq 3$ , with coefficient height  $\hat{h} = \log \max(1, |a_0|, \dots, |a_n|)$ , and  $f(x) = y^2$  in integers, then

$$(35) \quad \max\{h(x), h(y)\} \leq (4n)^{4096n^4} \exp(50n^4 \hat{h}), \quad h(t) = \log \max(1, |t|).$$

**Proposition 8.1** (Two effective constraints). *Every packet (32)–(33) satisfies*

$$(36) \quad H \leq \exp(4300p^5), \quad H \leq 2C_0D^{4050}.$$

*Proof.* First put  $f_p(t) = 4T_p(t) + 5$ . The relation in (33) gives  $f_p(X) = (2b+3)^2$ . The identities  $T'_p = pU_{p-1}$  and  $T_p^2 - 1 = (t^2 - 1)U_{p-1}^2$  show that a repeated root of  $f_p$  would have  $T_p = \pm 1$  and  $T_p = -5/4$ , a contradiction. If  $M_n$  is the sum of the absolute values of the coefficients of  $T_n$ , then  $M_{n+2} \leq 2M_{n+1} + M_n$  and  $M_0 = M_1 = 1$ . Induction gives  $M_n \leq 3^n$ . Thus every coefficient of  $f_p$  has absolute value at most  $4 \cdot 3^p + 5 \leq 3^{p+2}$ , and  $\hat{h} \leq (p+2) \log 3 \leq 4p$ . Since  $\log(4p) \leq p$  for  $p \geq 3$ , (35) implies

$$H \leq \log(2b+3) \leq \exp\left(4096p^4 \log(4p) + 50p^4 \hat{h}\right) \leq \exp(4300p^5).$$

For the second estimate, set  $x = b+1$  and  $Y = Duvr_0$ . The first three equations in (32) give

$$(37) \quad Y^2 = D(x^3 - x).$$

The cubic  $D(t^3 - t)$  is squarefree and has coefficient height  $\log D$ . Equation (35), with  $n = 3$ , gives

$$\log(b+1) \leq 12^{4096 \cdot 3^4} \exp(50 \cdot 3^4 \log D) = C_0 D^{4050}.$$

Finally  $b+2 \leq (b+1)^2$  for  $b \geq 1$ , proving the second bound.  $\square$

**Proposition 8.2** (Descent to the squarefree parameter). *Write  $D = d\tau^2$ , where  $d$  is a positive squarefree integer and  $\tau \geq 1$  is an integer. Then every packet satisfies*

$$H \leq 2C_0d^{4050}.$$

*Proof.* Equation (37) implies  $\tau^2 \mid Y^2$ . Comparing prime valuations gives  $\tau \mid Y$ . Write  $Y = \tau y$  and cancel the nonzero factor  $\tau^2$  to obtain

$$y^2 = d(x^3 - x), \quad x = b+1.$$

The polynomial  $d(t^3 - t)$  has distinct roots and coefficient height  $\log d$ . Equation (35) gives  $\log(b+1) \leq C_0d^{4050}$ , and again  $H \leq 2 \log(b+1)$ .  $\square$



Combining (34) and (36) gives

$$(38) \quad p \geq \left( \frac{\log H}{4300} \right)^{1/5}, \quad L < 4 \cdot 4300^{1/5} \frac{H}{(\log H)^{1/5}},$$

$$(39) \quad D \geq \left( \frac{H}{2C_0} \right)^{1/4050}, \quad p(\log H - K_0) < 16200H.$$

For example, the first upper bound follows by multiplying the lower bound for  $p$  by  $L > 0$  and applying  $pL < 4H$ . For the final inequality use  $\log H \leq K_0 + 4050 \log D$  and  $\log D < L$ . In a form avoiding fractional powers, the former calculation is

$$(\log H)L^5 \leq 4300(4H)^5.$$

These inequalities alone are compatible with unbounded height: they would force both the index and the squarefree kernel to grow, while  $\log(D+1)/H$  tends to zero. The following argument supplies the missing absolute index bound and excludes such an unbounded sequence in this specified family. It does not freeze a field or an index in advance.

**8.1. An absolute index bound from normalized logarithmic forms.** The preceding effective height estimate can be combined with a different logarithmic-form argument to prove finiteness, uniformly in the index. For this argument retain only positive integers  $b, r_0, s, X$  and an integer  $p \geq 3$  satisfying

$$(40) \quad b+2 = 3r_0^2, \quad b+3 = s^2, \quad Z = b^2 + 3b + 1 = T_p(X), \quad X > 1.$$

Primality of  $p$ , the endpoint kernels  $A, B$ , and their congruences are not needed. Put

$$\varepsilon = 2 + \sqrt{3}, \quad \eta = s + r_0\sqrt{3}, \quad \delta = X + \sqrt{X^2 - 1}, \quad W = Z + \sqrt{Z^2 - 1}, \quad H = \log(b+2).$$

The Pell equation gives  $\eta = \varepsilon^m$  for an integer  $m \geq 1$ . The Chebyshev identity gives  $W = \delta^p$ ;  $\delta$  need not be a fundamental unit. In fact  $b \geq 22$ : below 22 the fixed Pell equations leave only  $b = 1, r_0 = 1, s = 2$ , whereas  $Z = 5 < T_p(X)$  for  $p \geq 3, X \geq 2$ . Here  $T_p(X) \geq X^p$  follows by induction from its recurrence and  $T_{n+1}(X) \geq T_n(X) > 0$ .

We use Matveev's original Corollary 2.3 [20], with the normalized coefficient parameter in his equation (1.3). For a real number field of degree  $d$ , algebraic numbers  $\alpha_i > 1$ , and a nonzero form  $\Lambda = \sum_{i=1}^n b_i \log \alpha_i$  with integral coefficients, take

$$A_i \geq \max\{d\mathbf{h}(\alpha_i), |\log \alpha_i|, 0.16\}, \quad B_* = \max \left\{ 1, \max_i \frac{|b_i| A_i}{A_n} \right\}, \quad \Omega = \prod_i A_i.$$

Then

$$(41) \quad \log |\Lambda| > -C_1(n, 1)d^2\Omega \log(ed) \log(eB_*), \quad C_1(n, 1) \leq 2^{6n+20}.$$

Unlike Theorem 2.1 of that source, this corollary does not require the logarithms to be independent. It also does not require  $A_n$  to be the largest parameter. Both points matter when  $\delta$  varies.

**Lemma 8.3.** *For (40),*

$$1 < \frac{\eta^4}{8W} < 1 + \frac{2}{b}, \quad W > \eta^2, \quad \log W < 3H.$$

*Consequently the real linear form*

$$(42) \quad \Lambda = 4m \log \varepsilon - 3 \log 2 - p \log \delta = \log \frac{\eta^4}{8W}$$

*satisfies  $0 < \Lambda < 2/b$  and  $-\log \Lambda > H/2$ .*

*Proof.* For  $t > 1$ , write  $f(t) = t + \sqrt{t^2 - 1}$ , so  $2t - 1 < f(t) < 2t$ . The Pell identities give

$$\eta^2 = f(2b+5), \quad \eta^4 = f(8b^2 + 40b + 49), \quad W = f(Z).$$

Thus

$$\eta^4 > 16b^2 + 80b + 97 > 16Z > 8W,$$

and

$$\frac{\eta^4}{8W} - 1 < \frac{32b + 90}{16b^2 + 48b + 8} < \frac{2}{b}.$$

The final comparison is equivalent to  $32b^2 + 90b < 32b^2 + 96b + 16$ . Also  $W > 2b^2 + 6b + 1 > 4b + 10 > \eta^2$  for  $b \geq 22$ . Finally  $W < 2Z < 2(b+2)^2$  gives  $\log W < \log 2 + 2H < 3H$ . The strict inequality  $\log(1+u) < u$  for  $u > 0$  proves the asserted bounds for  $\Lambda$ . Since  $(b/2)^2 > b+2$ ,  $-\log \Lambda > \log(b/2) > H/2$ .  $\square$

**Theorem 8.4** (An absolute index bound). *Every tuple (40) satisfies  $p < 2^{59}$ .*

*Proof.* Use the real field  $K = \mathbb{Q}(\sqrt{3}, \sqrt{X^2 - 1})$  of degree  $d \leq 4$ , and in this order choose

$$(\alpha_1, \alpha_2, \alpha_3) = (\varepsilon, 2, \delta), \quad (b_1, b_2, b_3) = (4m, -3, -p),$$

$$A_1 = 2 \log \varepsilon, \quad A_2 = 4 \log 2, \quad A_3 = 2 \log \delta.$$

For integer  $X \geq 2$ ,  $X^2 - 1$  lies strictly between two consecutive squares. Hence  $\delta$  is a quadratic algebraic integer with minimal polynomial  $t^2 - 2Xt + 1$  and conjugate  $\delta^{-1}$ . The absolute heights are

$$h(\varepsilon) = \frac{1}{2} \log \varepsilon, \quad h(2) = \log 2, \quad h(\delta) = \frac{1}{2} \log \delta.$$

These identities,  $d \leq 4$ , and  $\delta \geq \varepsilon$  verify all the height and 0.16 conditions in (41). The fields may coincide; no independence assertion is used.

By Lemma 8.3, the normalized coefficients obey

$$\frac{4mA_1}{A_3} = \frac{4p \log \eta}{\log W} < 2p, \quad \frac{3A_2}{A_3} = \frac{6 \log 2}{\log \delta} < 4 \leq 2p.$$

For the latter strict inequality use  $\delta^2 \geq \varepsilon^2 > 8$ . The remaining entries are  $p$  and one, so  $B_* \leq 2p$ . Using the same chosen parameters, rather than changing them when estimating  $B_*$ , gives

$$\Omega = 16 \log \varepsilon \log 2 \frac{\log W}{p} < 96 \frac{H}{p}.$$

Apply (41), with  $C_1(3, 1) \leq 2^{38}$ ,  $d^2 \leq 16$ , and  $\log(ed) < 3$ . Comparison with the upper approximation gives

$$\frac{H}{2} < -\log \Lambda < 2^{38} \cdot 16 \cdot 3 \cdot 96 \frac{H}{p} \log(2ep).$$

Cancel the positive  $H$  and multiply by  $p$  to obtain

$$(43) \quad p < 9 \cdot 2^{48} \log(2ep) < 2^{52} \log(2ep).$$

With  $t = p/2^{52} > 0$ , this reads  $t < 53 \log 2 + 1 + \log t$ . The elementary inequality  $\log t \leq t/2$  gives  $t < 2(53 \log 2 + 1) < 108 < 128$ . Thus  $p < 2^{59}$ , independently of the height and all varying fields.  $\square$

**Corollary 8.5** (Effective finiteness of the specified family). *The tuples (40) form a finite set. Every such tuple satisfies*

$$(44) \quad H < \exp(4300 \cdot 2^{295}), \quad b + 2 < \exp(\exp(4300 \cdot 2^{295})).$$

*In particular the more restrictive residual (32)–(33) is effectively finite.*

*Proof.* The first part of the proof of Proposition 8.1 applies to  $4T_p(X) + 5 = (2b + 3)^2$  for every integer  $p \geq 3$ , giving  $H \leq \exp(4300p^5)$ . Combine it with Theorem 8.4 and exponentiate. The resulting bound leaves finitely many positive integers  $b$ . The square equations determine  $r_0, s$ , the index is bounded, and  $X^p \leq Z$  bounds  $X$ . If the endpoint data are also retained, then  $A \mid b$ ,  $B \mid b + 1$ , and their positive square multipliers have only finitely many possibilities.  $\square$

These bounds are effective but far beyond feasible enumeration. They prove finiteness without listing the exceptional tuples. They improve a fixed-index finiteness statement: the same absolute bound applies while both the index and the quadratic field vary.

**Corollary 8.6** (The product Pell unit outside the finite set). *Suppose only that  $b, r_0, s$  are positive integers with  $b + 2 = 3r_0^2$  and  $b + 3 = s^2$ . Put  $Z = b^2 + 3b + 1$  and write  $Z^2 - 1 = D_0 y^2$  with  $D_0 > 1$  squarefree and  $y$  a positive integer. Outside the bound in (44),  $Z + y\sqrt{D_0}$  is the fundamental positive norm-one unit of the integral Pell group  $\mathbb{Z}[\sqrt{D_0}]$ .*

*Proof.* Write  $Z + y\sqrt{D_0} = \varepsilon_{D_0}^k$ ,  $k \geq 1$ . If  $k = 2j$ , then  $Z = T_2(T_j(X_0))$  for the integer first coordinate  $X_0 > 1$  of  $\varepsilon_{D_0}$ . Thus, with  $Y = T_j(X_0)$ ,

$$2Y^2 = Z + 1 = (b + 1)(b + 2) = 3(b + 1)r_0^2.$$

Since  $3 \nmid b + 1$ , the right side has odd 3-adic valuation and the left side has even 3-adic valuation, a contradiction. Therefore  $k$  is odd. If  $k > 1$ , it is at least three and (40) holds with  $p = k$ ,  $X = X_0$ . Corollary 8.5 bounds  $b$ , proving the assertion.  $\square$

This last statement concerns the integral Pell group, without identifying it with the unit group of a larger maximal order. A fundamental moving Pell unit can still have a large regulator relative to its discriminant. Neither finiteness of the nonfundamental cases nor the absolute index bound gives the radical estimate (1), and no reduction of arbitrary *abc* triples to (40) has been established.

**8.2. A stronger bound from a fixed-norm element.** For the height and kernel estimates in this subsection use only (32), and impose the additional conditions that  $A, B$  are squarefree and

$$A \equiv 22 \pmod{24}, \quad B \equiv 23 \pmod{24}.$$

They specify the squarefree branch under consideration; the more general packet estimates above did not require them. No Chebyshev decomposition is needed except in the displayed consequence for  $p$ . In particular  $b \geq 22$ ,  $A, B > 1$ ,  $3 \nmid AB$ ,  $\gcd(A, B) = 1$ , and  $D = 3AB$  is squarefree. Let  $R_A, R_B$  denote the regulators of the maximal real quadratic orders.

We use two general tools from the proof of the new result [23, Theorem 2.4 and Lemma 2.5], without applying its irreducible-cubic theorem to a split polynomial. First, in a totally real degree-four field, for a multiplicative group generated modulo torsion by three independent elements  $g_1, g_2, g_3$  and  $g \neq 1$  in that group,

$$(45) \quad -\log |1 - g| \leq C_* \log \max\{2, h(g)\} \prod_{i=1}^3 \max\{1, h(g_i)\},$$

where  $C_* > 0$  is effective and absolute. Second, for an algebraic integer  $\omega$  of absolute norm  $q > 1$  in a real quadratic field  $K$ , the group  $\mathcal{O}_K^\times \langle \omega \rangle$  has independent elements  $\xi_1, \xi_2$  generating a subgroup modulo torsion of index  $1 \leq N \leq 2$  such that

$$(46) \quad \omega^N = \zeta \xi_1^{e_1} \xi_2^{e_2}, \quad \zeta \in \{\pm 1\}, \quad h(\xi_1)h(\xi_2) \leq \frac{3}{8} R_K \log q.$$

Here the set  $S$  consists of both real places and no finite place, so the all-or-none condition in Lemma 2.5 is satisfied and  $q' = q$ . The small-generator estimate follows from the exterior-product theorem of Akhtari–Vaaler [1, Theorem 1.2]; its joint use of the unit and nonunit generators is what saves a regulator factor. Finally Landau’s fixed-degree estimate gives an effective absolute  $C_Q$  with

$$(47) \quad R_A \leq C_Q \sqrt{A} \log(2A), \quad R_B \leq C_Q \sqrt{B} \log(2B).$$

**Theorem 8.7.** *There is an effective absolute constant  $C_A$  such that every packet in this subsection satisfies*

$$H \leq C_A \sqrt{A} [\log(2A)]^2.$$

*Proof.* Put  $\varphi = (1 + \sqrt{5})/2$  and  $\rho = \frac{1}{2} \log \varphi$ . Every nonzero algebraic number of degree at most two which is not a root of unity has height at least  $\rho$ . Indeed a rational such number has height at least  $\log 2$ . For a quadratic number, its primitive minimal polynomial has Mahler measure at least two if the leading or constant coefficient has absolute value at least two. Otherwise it is a quadratic unit. For norm minus one a nonzero integral trace gives a root of absolute value

at least  $\varphi$ ; trace zero is reducible. For norm plus one, traces of absolute value at most two give roots of unity or reducibility, and the other traces give a root of absolute value at least  $\varphi^2$ . The formula  $2h = \log M$  proves the height lower bound.

In  $K = \mathbb{Q}(\sqrt{A})$  take

$$\omega = s + u\sqrt{A} = \sqrt{b+3} + \sqrt{b}, \quad \eta = s + r_0\sqrt{3} = \sqrt{b+3} + \sqrt{b+2}.$$

Their respective quadratic norms are three and one. The latter is a positive power of  $\varepsilon_3 = 2 + \sqrt{3}$ . Apply (46) with  $q = 3$ . The field  $F = \mathbb{Q}(\sqrt{A}, \sqrt{3})$  has degree four, since  $A > 1$  is squarefree and  $A \neq 3$ . Define  $\gamma = \omega/\eta$ . At the chosen real embedding,

$$0 < \gamma < 1, \quad 1 - \gamma = \frac{2}{(\sqrt{b+2} + \sqrt{b})(\sqrt{b+3} + \sqrt{b+2})} < \frac{1}{2b}.$$

As  $1 \leq N \leq 2$ , it follows that  $0 < 1 - \gamma^N < 1/b$ .

The group  $\langle -1, \xi_1, \xi_2, \varepsilon_3 \rangle$  has rank three modulo torsion and contains  $\gamma^N$ . Independence follows because  $K \cap \mathbb{Q}(\sqrt{3}) = \mathbb{Q}$ , and a rational power of  $\varepsilon_3$  must equal its conjugate, forcing that power to be zero. The other two generators are independent by construction. Moreover  $h(\varepsilon_3) < 1$  and  $\max(1, h(\xi_i)) \leq h(\xi_i)/\rho$ , giving

$$\prod_i \max(1, h(g_i)) \leq \frac{3 \log 3}{8\rho^2} R_A.$$

Both conjugates of  $\omega, \eta$  are positive. Their smaller conjugates are  $3/\omega < 1$  and  $1/\eta < 1$ , so their global heights are  $\frac{1}{2} \log \omega$  and  $\frac{1}{2} \log \eta$ . Thus

$$h(\gamma^N) \leq 2(h(\omega) + h(\eta)) < \log(4(b+3)) < H + 2 \leq 3H.$$

Applying (45) gives  $\log b \ll R_A \log(3H)$ , with an effective absolute constant. Since  $H \leq \log b + \log 3$ , we have  $H \leq M \log(3H)$  for  $M = C_1 \max(1, R_A) \geq 1$  and an absolute  $C_1$ .

For any positive  $X, M$  with  $X \leq M \log(3X)$ , one has

$$(48) \quad X \leq 2M \log(3M).$$

To prove it, write  $t = X/M$  and use  $\log t \leq t/2$ , which follows from  $\log(t/2) \leq t/2 - 1$  and  $\log 2 < 1$ . Then  $t \leq \log(3M) + t/2$ , proving (48). Equation (47) bounds  $M$  by  $C_2 \sqrt{A} \log(2A)$  and  $\log(3M)$  by  $C_3 \log(2A)$ . Applying (48) proves the theorem.  $\square$

Replacing  $\omega$  by  $s + v\sqrt{B}$ , of norm two, gives the analogous bound with  $\sqrt{B} \log^2(2B)$ . For this endpoint, however, a stronger published input is available and was already identified in the repository. The unit

$$\varepsilon_B = (b+2) + vs\sqrt{B}$$

is the fundamental positive norm-one unit: its first coordinate is  $3r_0^2$ , so Bennett–Walsh [5, Theorem 1.2] says that its unit index is the first index at which three divides the first Pell coordinate. If the fundamental first coordinate is 1 or  $-1$  modulo three, the Chebyshev recurrence gives only 1 or alternating signs modulo three, respectively. A first coordinate divisible by three therefore forces the fundamental first coordinate itself to be divisible by three, and the index is one. Since  $B \equiv 3 \pmod{4}$ , the order  $\mathbb{Z}[\sqrt{B}]$  is maximal and a norm-minus-one unit is impossible modulo four. Consequently

$$(49) \quad H < R_B \leq C_Q \sqrt{B} \log(2B).$$

We use this established endpoint result, rather than presenting it as a new consequence of the recent preprint.

**Corollary 8.8.** *There is an effective absolute  $K \geq 1$  such that these packets satisfy*

$$(50) \quad H^4 \leq KD[\log(2D)]^6.$$

*In particular,*

$$(51) \quad D \geq \frac{H^4}{K5^6(\log H)^6}, \quad \log D \geq 4 \log H - 6 \log \log H - \log(K5^6).$$

When the latter lower bound is positive,

$$p < \frac{4H}{4 \log H - 6 \log \log H - \log(K5^6)}.$$

*Proof.* Square Theorem 8.7 and (49) and multiply. Since  $AB = D/3$  and  $A, B \leq D$ , the result is (50). If  $D \geq H^4$ , the first inequality in (51) is immediate because  $H > 2$  and  $K \geq 1$ . Otherwise  $\log(2D) \leq 5 \log H$ , giving that inequality from (50). Taking logarithms proves the second one. Finally use  $p \log(D+1) < 4H$ .  $\square$

The absolute index bound already excludes an unbounded sequence with  $p \geq 3$ . The kernel estimate (50) remains useful without a Chebyshev decomposition, including cases where the product Pell unit is fundamental. Its polynomial dependence on  $H$  does not supply the radical bound required for *abc*.

**Proposition 8.9** (A general split-cubic estimate). *For positive squarefree  $d$ , an integral solution of  $y^2 = d(x^3 - x)$  with  $x \geq 2$  satisfies*

$$\log x \ll d^{2/3} [\log(2d)]^3$$

*with an effective absolute implied constant.*

*Proof.* Write  $x-1 = a_0 u^2$ ,  $x = b_0 v^2$ ,  $x+1 = c_0 w^2$  with positive squarefree coefficients. Adjacent coefficients are coprime and  $g = \gcd(a_0, c_0) \in \{1, 2\}$ , so  $a_0 b_0 c_0 = g^2 d \leq 4d$ . No two coefficients are equal: equality for adjacent ones would force positive squares to differ by one; equality at the ends forces the common coefficient to divide two and gives the same impossibility or a difference of squares equal to two. The quadratic fields of the three products  $a_0 b_0, b_0 c_0, a_0 c_0$  are therefore distinct.

The three positive norm-one units

$$U = 2x - 1 + 2uv\sqrt{a_0 b_0}, \quad V = 2x + 1 + 2vw\sqrt{b_0 c_0}, \quad W = x + uw\sqrt{a_0 c_0}$$

satisfy  $1 < V/U, V/(2W), 2W/U < 1 + 2/x$ . Indeed  $2t - 1 < t + \sqrt{t^2 - 1} < 2t$  for  $t > 1$  gives these inequalities directly at  $t = 2x - 1, 2x + 1, x$ . Each ratio is generated by two quadratic fundamental units and two; two may also be adjoined for the first ratio. Distinct quadratic fields give independence, and two is an independent nonunit. The global height of each ratio is at most  $\log x + O(1)$ . Choose the pair whose radicands share  $m = \min(a_0, b_0, c_0)$ . Landau's bound gives a product of regulators at most

$$C \sqrt{a_0 b_0 c_0 m} [\log(2a_0 b_0 c_0)]^2 \leq C (a_0 b_0 c_0)^{2/3} [\log(2a_0 b_0 c_0)]^2.$$

Passing to squarefree radicands can only decrease the discriminant bound. Apply (45) to the selected ratio and then (48), adjusting absolute constants for bounded  $x$ . The latter step adds one logarithmic power. Using  $a_0 b_0 c_0 \leq 4d$  proves the claim.  $\square$

## 9. APPLICABILITY OF RECENT ELLIPTIC-CURVE ESTIMATES

**9.1. A denominator lower bound.** For the Frey curve  $E : y^2 = x(x-a)(x+b)$  attached to an *abc* triple, let  $Q = a^2 + ab + b^2$ . Its invariant is

$$j_E = \frac{256Q^3}{(abc)^2}.$$

Let  $n, d$  be the positive numerator and denominator in lowest terms.

**Proposition 9.1.** *For every such triple,*

$$(52) \quad c^4 \leq 1024d, \quad n \leq 256c^6.$$

*Proof.* Primitivity implies  $\gcd(Q, abc) = 1$ . Indeed, reduction modulo a prime dividing any one of  $a, b, c$  leaves in  $Q$  the square of a nonzero other coordinate. Therefore  $g = \gcd(256Q^3, (abc)^2)$  divides 256, and  $(abc)^2 = gd \leq 256d$ . Since  $(a-1)(b-1) \geq 0$ , we have  $ab \geq c-1$ ; as  $c \geq 2$ , this gives  $2ab \geq c$ . Squaring  $2abc \geq c^2$  proves  $c^4 \leq 4(abc)^2 \leq 1024d$ . Finally  $Q = c^2 - ab \leq c^2$ , so  $n \leq 256Q^3 \leq 256c^6$ .  $\square$

Pasten's 2026 small-denominator theorem [24, Theorem 1.2] applies under a bound  $\text{den}(j_E) \leq A_0(\log \text{num}(j_E))^{B_0}$  with fixed positive  $A_0, B_0$ . For Frey curves, (52) would imply

$$c^4 \leq 1024A_0(\log 256 + 6 \log c)^{B_0}.$$

A positive power of  $c$  dominates every fixed power of its logarithm. Thus only bounded  $c$ , and consequently finitely many positive triples, can satisfy this hypothesis for fixed  $A_0, B_0$ . This is a precise restriction on direct applicability, not a counterexample to the theorem or to other Szpiro strategies.

**9.2. The August 2026 Mordell estimate.** The preprint [23, Theorem 1.3], submitted August 24, 2026, gives an effective bound for  $y^2 = x^3 + k$ ,  $k \neq 0$ , using

$$\underline{k} = \prod_{q|k} q^{\min(2, v_q(k))}.$$

The stated estimate is

$$(53) \quad \log \max(|x|, |y|) \ll \underline{k}^{1/2} \log^2(2k) \log(2|k|) \log(\underline{k} \log(3|k|)).$$

We record this new result without treating its publication date as a substitute for checking its hypotheses or proof.

Directly substituting  $c_6^2 = c_4^3 - 1728\Delta$  and  $\Delta = 16(abc)^2$  gives  $k = -27648(abc)^2$ . Then  $\underline{k}^{1/2} \ll r$ , while both  $\log(2|k|)$  and  $\log \max(|c_4|, |c_6|)$  are comparable with  $h$ . The resulting right-hand side has the shape

$$r \log^2(2r) h \log(2r \log(3c)).$$

The height remains a multiplicative factor on the right; cancellation does not yield an upper bound for  $c$  in terms of  $r$ . The integral- $j$  result in Theorem 1.6 of the same paper concerns the height of the *curve*, and cannot be substituted for a point-height bound.

There is a more primitive Mordell point, but it preserves the same height factor in this substitution.

**Proposition 9.2.** Put  $m = abc/2$  and  $Y = (a - b)(2a + b)(a + 2b)/2$ . These are integers, and

$$Y^2 = Q^3 - 27m^2, \quad \gcd(Q, m) = 1, \quad c_4 = 4^2Q, \quad c_6 = 4^3Y.$$

There is no integer  $z \geq 2$  with  $z^2 \mid Q$  and  $z^3 \mid Y$ . The associated binary cubic is split over  $\mathbb{Z}$ :

$$U^3 - 3QUV^2 + 2YV^3 = (U - (a - b)V)(U - (a + 2b)V)(U + (2a + b)V).$$

It remains reducible after every change of variables in  $\text{GL}_2(\mathbb{Z})$ .

*Proof.* At least one of  $a, b, c$  is even, so  $m$  is integral. If one of  $a, b$  is even then one of  $2a + b, a + 2b$  is even; if both are odd then  $a - b$  is even. This proves integrality of  $Y$ . Expansion gives

$$4Q^3 - ((a - b)(2a + b)(a + 2b))^2 = 27(abc)^2$$

and the asserted cubic factorization. The standard invariant formulas give the two scalings. Coprimality follows from  $\gcd(Q, abc) = 1$ . If the proposed  $z$  existed, then  $z^6 \mid 27m^2$  while  $\gcd(z, m) = 1$ ; thus  $z^6 \mid 27$ , impossible for  $z \geq 2$ . An invertible linear change of variables takes each nonzero linear factor to another nonzero linear factor, preserving reducibility.  $\square$

The normalized parameter is therefore  $k = -27m^2$  and its prime support is still that of  $3m$ . Removing the fixed powers of four does not remove the factor comparable with  $h$  on the right of (53). The split cubic also does not meet an irreducible Thue-form hypothesis.

There is also a strict obstruction to one possible transfer from (37). With  $U = Dx$  and  $V = DY$ , it becomes  $E_D : V^2 = U^3 - D^2U$ , of  $j$ -invariant 1728.

**Proposition 9.3.** For  $D, k \neq 0$ , there is no nonconstant algebraic morphism over  $\overline{\mathbb{Q}}$  from  $E_D$  to the nonsingular Mordell curve  $M_k : v^2 = u^3 + k$ .



*Proof.* Their geometric endomorphism algebras are respectively  $\mathbb{Q}(i)$  and  $\mathbb{Q}(\sqrt{-3})$ : the automorphisms  $(x, y) \mapsto (-x, iy)$  and  $(u, v) \mapsto (\zeta_3 u, v)$  give these quadratic actions, and in characteristic zero the endomorphism algebra of an elliptic curve has dimension at most two over  $\mathbb{Q}$ . A nonconstant morphism, translated to send the origin to the origin, is an isogeny. An isogeny induces an isomorphism of rational endomorphism algebras by conjugation. The two imaginary quadratic fields are not isomorphic, a contradiction; see also [25, Chapters III and VI].  $\square$

The proposition excludes this morphism strategy, even if  $k$  depends on  $D$ . It does not exclude other Diophantine encodings or extending the new logarithmic-form methods to a different family.

## 10. THREE AUXILIARY POINTS AND THE REMAINING ENDPOINT COST

The constructions here apply to every positive primitive  $abc$  triple. Write its canonical cube decompositions as

$$(54) \quad a = Au^3, \quad b = Bv^3, \quad c = Cw^3,$$

where  $A, B, C$  are positive cube-free integers. These coefficients, and the three original endpoints, are pairwise coprime within their respective triples. Put

$$R_0 = \text{rad}(ABC), \quad R_3 = \prod_{\substack{p|abc \\ 3|v_p(abc)}} p, \quad H = \log c.$$

Thus  $\text{rad}(abc) = R_0 R_3$  and  $\gcd(R_0, R_3) = 1$ . The subscript on  $R_0$  denotes the support left after removing cubes, not the primes whose exponents are divisible by three.

**10.1. Actual integral points and their prime costs.** For each omitted endpoint  $n_i$ , define the following integers:

| $i$ | $n_i$ | $d_i$ | $r_i$ | $x_i$    | $y_i$         |
|-----|-------|-------|-------|----------|---------------|
| $a$ | $a$   | $vw$  | $BC$  | $4vwBC$  | $4BC(a + 2b)$ |
| $b$ | $b$   | $uw$  | $AC$  | $4uwAC$  | $4AC(2a + b)$ |
| $c$ | $c$   | $uv$  | $AB$  | $-4uvAB$ | $4AB(a - b)$  |

Set  $k_i = 16(n_i r_i)^2$ .

**Proposition 10.1.** *Every row gives an integral point  $y_i^2 = x_i^3 + k_i$  with  $|x_i|^3 \geq 32c$ . If  $\rho_i = \text{rad}(2n_i r_i)$ , then*

$$(55) \quad \rho_a \rho_b \rho_c = 4^{1_{3|v_2(abc)}} R_0^3 R_3, \quad \min_i \rho_i \leq 2^{2/3} R_0 R_3^{1/3}.$$

*Proof.* For the first row use  $(a + 2b)^2 - a^2 = 4bc$  and  $bc = (vw)^3 BC$ ; the second is the same identity with  $a, b$  interchanged. For the last,  $(a - b)^2 - c^2 = -4ab$  gives the negative sign of  $x_c$ . The three values of  $|x_i|^3$  are respectively  $64bc(BC)^2$ ,  $64ac(AC)^2$ , and  $64ab(AB)^2$ . The first two exceed  $32c$ , and the last is at least  $32c$  because  $ab \geq c - 1 \geq c/2$ .

An odd prime in  $R_0$  occurs in all three  $n_i r_i$ : once in the omitted endpoint and twice in the coefficient products. A prime in  $R_3$  occurs only in its own omitted endpoint. The prime 2 divides every  $\rho_i$ , and exactly one endpoint is even. If its exponent is divisible by three this adds a factor 4 to the preceding product; otherwise no correction is needed. This proves the exact identity; the minimum is at most the geometric mean.  $\square$

These are auxiliary curves of  $j$ -invariant zero. Their standard 3-isogeny is

$$(56) \quad (x, y) \mapsto \left( \frac{x^3 + 4k}{x^2}, \frac{y(x^3 - 8k)}{x^3} \right), \quad E_k \longrightarrow E_{-27k}.$$

On the unextracted omit- $b$  point with  $k = 16(abc)^2$ , it gives  $(4Q, 4T)$ , where

$$Q = a^2 + ab + b^2, \quad T = (a - b)(2a + b)(a + 2b).$$

These are  $(c_4/4, c_6/8)$  for the original Frey curve. The assertion is an identity involving its invariants; the Frey curve itself has not become isogenous to a  $j = 0$  curve. After cube extraction, the same image is generally rational rather than integral.

**10.2. A small field discriminant with an exact growing order index.** Form the monic cubic

$$(57) \quad G_i(Z) = Z^3 - 3x_iZ + 2y_i, \quad \text{disc } G_i = -108k_i = -1728(n_i r_i)^2.$$

For the omit- $b$  row, put  $\alpha = \sqrt[3]{A^2C}$  and  $\beta = \sqrt[3]{AC^2} = \alpha^2/A$ . Then  $\tau_b = -2(u\alpha + w\beta)$  is a root. The omit- $a$  row uses  $(B, C, v, w)$  instead, while the omit- $c$  row has root  $\tau_c = -2(u\sqrt[3]{A^2B} - v\sqrt[3]{AB^2})$ .

If the selected product  $r_i$  exceeds one, its pure cubic is irreducible. At a prime of either cube-free coefficient, the valuation of the radicand is not divisible by three. Independence of  $1, \alpha, \alpha^2$  then makes  $\tau_i$  nonrational, so it generates the same degree-three field  $K_i$ . In particular the root fields are

$$K_a = \mathbb{Q}(\sqrt[3]{B^2C}), \quad K_b = \mathbb{Q}(\sqrt[3]{A^2C}), \quad K_c = \mathbb{Q}(\sqrt[3]{A^2B}),$$

with signature  $(1, 1)$ . Reducible rows are exactly those with  $r_i = 1$ .

**Proposition 10.2.** *In an irreducible row, put  $S_i = \text{rad}(r_i)$ . There is an order  $\mathcal{O}_i \subset K_i$  with*

$$(58) \quad [\mathcal{O}_i : \mathbb{Z}[\tau_i]] = 8n_i, \quad |\Delta_{K_i}| \mid 27S_i^2.$$

*More precisely, there is a larger order  $\mathcal{O}'_i$  such that*

$$(59) \quad [\mathcal{O}_{K_i} : \mathbb{Z}[\tau_i]] = 8n_i(r_i/S_i)[\mathcal{O}_{K_i} : \mathcal{O}'_i].$$

*Proof.* Write the two selected coefficients as  $E, F$ , their cube bases as  $h, j$ , and  $r = EF$ . The lattice  $\mathcal{O} = \mathbb{Z}\{1, \alpha, \beta\}$  is an integral order, because

$$\alpha^2 = E\beta, \quad \beta^2 = F\alpha, \quad \alpha\beta = r.$$

Its trace matrix is  $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 3r \\ 0 & 3r & 0 \end{pmatrix}$ , and hence its discriminant is  $-27r^2$ . The coordinate matrices of  $1, \tau, \tau^2$  in this basis are

$$M_+ = \begin{pmatrix} 1 & 0 & 8hjr \\ 0 & -2h & 4Fj^2 \\ 0 & -2j & 4Eh^2 \end{pmatrix}, \quad M_- = \begin{pmatrix} 1 & 0 & -8hjr \\ 0 & -2h & 4Fj^2 \\ 0 & 2j & 4Eh^2 \end{pmatrix}.$$

Their determinants are  $8(Fj^3 - Eh^3)$  and  $-8(Eh^3 + Fj^3)$ , respectively. The absolute value is  $8n_i$  in the corresponding row. Since  $\tau$  is an integral generator of a cubic field, this gives the first equality of (58).

Write  $E = E_1E_2^2$ ,  $F = F_1F_2^2$  with the four factors squarefree and pairwise coprime. Put  $s = E_2F_1$ ,  $t = E_1F_2$ ,  $\theta = \sqrt[3]{st^2}$ , and  $\eta = \theta^2/t$ . The lattice  $\mathcal{O}' = \mathbb{Z}\{1, \theta, \eta\}$  is an order by  $\theta^2 = t\eta$ ,  $\eta^2 = s\theta$ ,  $\theta\eta = st$ . Its discriminant is  $-27(st)^2 = -27S_i^2$ . Furthermore  $\alpha = E_2\theta$ ,  $\beta = F_2\eta$ , so  $[\mathcal{O}' : \mathcal{O}] = E_2F_2 = r_i/S_i$ . The discriminant-index formula and multiplication of indices prove the remaining claims.  $\square$

Thus the small discriminant concerns the field, while the original monogenic order has an index containing the unbounded endpoint  $n_i$ . Replacing its polynomial discriminant by the field discriminant in a point-height argument would discard this factor.

**10.3. A full-family relative height estimate.**

**Theorem 10.3.** *There is an effective absolute constant  $C_0 > 0$  such that every positive primitive abc triple, and each of its three rows, satisfies*

$$(60) \quad H \leq C_0 S_i [\log(2S_i)]^2 L_i \log(3S_i L_i), \quad L_i = \log(2n_i r_i), \quad S_i = \text{rad}(r_i).$$

*Proof.* We use the stated unconditional Theorem 1.7 and the reduction in Section 4 of Pasten's 2026 preprint [23], together with its classical degree-three regulator bound. We keep the actual pure cubic root field, rather than bounding its regulator using disc  $G_i$ .

First suppose  $r_i > 1$  and put  $D = 1728(n_i r_i)^2$ . The reduction in that section supplies  $\gamma \in \mathrm{GL}_2(\mathbb{Z})$  such that the binary cubic  $F = G_i \circ \gamma$  has coefficient height at most  $\sqrt{D}$ . For  $(u_0, v_0) = \gamma^{-1}(1, 0)$  one has  $F(u_0, v_0) = 1$  and  $\gcd(u_0, v_0) = 1$ . The fractional linear change of root leaves the cubic field  $K_i$  unchanged. Theorem 1.7, applied with empty prime list, unit right side, and just the two archimedean places of  $K_i$ , gives, for  $X = \max(3, |u_0|, |v_0|)$ ,

$$(61) \quad \log X \ll \log(2D) + \mathcal{R}_i \log(2D) \log(2 + 2\mathcal{R}_i \log(2D)),$$

where  $\mathcal{R}_i$  is the ordinary regulator. The degrees and the place count are fixed, so the implied constant is absolute and effective. Hessian and Jacobian recovery in the same section gives

$$\log \max(2, |x_i|, |y_i|) \ll \log |k_i| + \log X.$$

These covariants recover the point, not just the auxiliary curve. Proposition 10.1, together with  $\log(2D) \leq 12L_i$  and  $\log |k_i| \leq 4L_i$ , therefore yields

$$H \ll \mathcal{R}_i^* L_i \log(3\mathcal{R}_i^* L_i), \quad \mathcal{R}_i^* = \max(1, \mathcal{R}_i).$$

Here all logarithm arguments are bounded away from one because  $L_i \geq \log 2$ .

The degree-three Landau bound, with class number at least one, and Proposition 10.2 imply

$$\mathcal{R}_i^* \ll S_i [\log(2S_i)]^2.$$

Substitution loses no further logarithmic power. Indeed, for  $B = \log(2S_i)$  and  $z = S_i L_i \geq \log 2$ , one has  $B \leq 2S_i$  and  $L_i \geq 1/2$ , hence  $S_i B^2 L_i \leq 16z^3$ . Thus  $\log(3C S_i B^2 L_i) \ll_C \log(3S_i L_i)$  for any fixed positive  $C$ . This proves (60) in the irreducible case.

If  $r_i = 1$ , the selected endpoints are cubes. In a plus row their difference is  $n_i = w^3 - h^3$  with  $w > h \geq 1$ . Thus  $n_i \geq w^2$  and  $H = 3 \log w \leq 3L_i/2$ . In the minus row  $n_i = c$ , so  $H \leq L_i$ . As  $S_i = 1$  and  $L_i \geq \log 2$ , enlarging one absolute constant proves the same bound in all reducible cases.  $\square$

Cube-freeness gives  $r_i \leq S_i^2$ , so  $L_i$  may be replaced on the right by  $\log(2n_i) + 2 \log S_i$ . Also

$$(62) \quad S_a S_b S_c = R_0^2, \quad \min_i S_i \leq R_0^{2/3}.$$

Choosing the omitted endpoint  $m = \min(a, b)$  instead gives

$$(63) \quad H \ll R_0 [\log(2R_0)]^2 M \log(3R_0 M), \quad M = \log(2m) + 2 \log R_0.$$

It is not legitimate to cancel an unknown height from the right:  $L_i$  may be comparable to  $H$ . For fixed bounded  $R_0$ , (63) does imply  $\log m \gg_{R_0} H / \log H$  along an unbounded sequence. A stronger fixed-support statement follows next, with an ineffective threshold.

**10.4. One common curve and its three torsion translates.** Put  $N = ABC$  and  $t_a = u, t_b = v, t_c = w$ . Since  $n_i r_i = N t_i^3$ , the points

$$Q_i = (X_i, Y_i) = (x_i/t_i^2, y_i/t_i^3)$$

all lie on the same nonsingular curve

$$(64) \quad E_N : Y^2 = X^3 + 16N^2.$$

Primitivity gives  $\gcd(t_i, d_i r_i) = 1$ , so their reduced  $X$ -denominators are exactly

$$(65) \quad \mathrm{den}(X_i) = (t_i / \gcd(t_i, 2))^2.$$

Indeed the numerator is  $\pm 4d_i r_i$ , and the only cancellation with  $t_i^2$  is  $\gcd(t_i^2, 4) = \gcd(t_i, 2)^2$ .

**Proposition 10.4.** *The point  $T_0 = (0, 4N)$  has order three on  $E_N$ , and*

$$Q_b + T_0 = Q_c, \quad Q_b - T_0 = -Q_a.$$

*Proof.* The horizontal tangent at  $T_0$  gives  $2T_0 = -T_0$ . The slopes from  $T_0$  and  $-T_0$  to  $Q_b$  are respectively

$$\frac{2Au^2}{vw}, \quad \frac{2Cw^2}{uv}.$$

Squaring the first and subtracting  $X_b$  gives  $-4ABuv/w^2 = X_c$ ; the ordinate from the line through  $T_0$  is  $4AB(a-b)/w^3 = Y_c$ . The second gives  $4BCvw/u^2 = X_a$  and ordinate  $-4BC(a+2b)/u^3 = -Y_a$ . These computations use only  $Au^3 + Bv^3 = Cw^3$  and positive denominators.  $\square$

Writing  $V = uvw$ , the three points  $Q_b, Q_c, -Q_a$  have the same image  $(4Q/V^2, 4T/V^3)$  under (56). Their changing denominators describe three translates of one rational point; an integral-point bound for a fixed curve does not bound them.

**Theorem 10.5** (Asymptotic balance for bounded residual support). *Fix  $R_* \geq 1$ . For every  $\varepsilon > 0$ , all sufficiently large positive primitive abc triples with  $R_0 \leq R_*$  satisfy*

$$\min(a, b) \geq c^{1-\varepsilon}.$$

*Equivalently, along any such sequence with  $c \rightarrow \infty$ ,  $\log \min(a, b)/\log c \rightarrow 1$ . The threshold may depend on  $R_*, \varepsilon$ ; the proof does not make it effective.*

*Proof.* For a fixed elliptic curve over  $\mathbb{Q}$ , Siegel's local approximation theorem, with the even function  $x$ , the point at infinity, and the real place, states that

$$(66) \quad \frac{\log \max(1, |x(P)|)}{h_x(P)} \rightarrow 0 \quad \text{as } h_x(P) \rightarrow \infty, \quad P \in E(\mathbb{Q}).$$

Here  $h_x(P) = \log \max(|A_x|, B_x)$  for  $x(P) = A_x/B_x$  in lowest terms,  $B_x > 0$ . This is [25, Theorem IX.3.1 and Example IX.3.3], not just finiteness of integral points. Since  $N = ABC \leq R_0^2 \leq R_*^2$ , only finitely many curves (64) occur. Formula (66) is uniform over this finite list.

By symmetry suppose  $a = m \leq b$ , and set  $g = \gcd(u, 2) \in \{1, 2\}$ . Equation (65) gives

$$X_a = A_x/B_x, \quad A_x = 4vwBC/g^2, \quad B_x = u^2/g^2.$$

Moreover  $X_a^3 = 64N^2bc/a^2 \geq 64$ , so  $X_a \geq 4$  and  $h_x(Q_a) = \log A_x$ . Direct substitution yields

$$\log A_x = \frac{1}{3}(\log b + \log c) + \frac{2}{3} \log(BC) + \log 4 - 2 \log g,$$

$$\log B_x = \frac{2}{3} \log a - \frac{2}{3} \log A - 2 \log g.$$

As  $c/2 \leq b < c$ , these imply

$$(67) \quad \frac{2}{3}H - \frac{1}{3} \log 2 \leq h_x(Q_a) \leq \frac{2}{3}H + \log 4 + \frac{2}{3} \log N,$$

$$(68) \quad 0 \leq H - \log a \leq \frac{3}{2} \log X_a + \frac{1}{2} \log 2.$$

Thus  $h_x(Q_a) \rightarrow \infty$  and is comparable to  $H$ , uniformly for bounded  $N$ . Equation (66) makes the right side of (68) equal to  $o(H)$ . This proves the stated ratio and hence the power inequality. If  $b < a$ , use  $Q_b$  instead. An infinite countersequence to the eventual assertion would contradict the same finite-list application of Siegel, so the quantifier is over all triples with  $R_0 \leq R_*$ .  $\square$

**10.5. An infinite fixed-support family and the exact limitation.** Bounded  $R_0$  alone does not imply bounded height.

**Proposition 10.6.** *There are infinitely many positive primitive abc triples of unbounded height with  $R_0 = 3$ .*

*Proof.* On  $Y^2 = X^3 - 34992$ , let  $P = (36, -108)$ . Doubling gives

$$2P = (252, 3996), \quad X(4P) = \left(\frac{882}{37}\right)^2 - 504 = \frac{87948}{1369}.$$

The final numerator has remainder 36 modulo 37, so  $4P$  is not integral. The Nagell–Lutz integrality theorem on this integral nonsingular short Weierstrass model implies that  $P$  has infinite order; see [26, Theorem 24.21]. The mutually inverse formulas

$$u = \frac{324 + Y}{6X}, \quad v = \frac{324 - Y}{6X}, \quad X = \frac{108}{u + v}, \quad Y = \frac{324(u - v)}{u + v}$$

identify its rational affine points with  $u^3 + v^3 = 9$ . No affine rational point has  $X = 0$ , and  $u + v = 0$  is impossible. Nonzero multiples of  $P$  therefore supply infinitely many rational solutions. Clear denominators to obtain

$$x^3 + y^3 = 9z^3, \quad z > 0, \quad \gcd(x, y, z) = 1.$$

Neither  $x$  nor  $y$  is zero, since 9 is not a rational cube. If  $3 \mid x$ , reduction modulo three forces  $3 \mid y$ , and then  $27 \mid 9z^3$  forces  $3 \mid z$ , a contradiction. The same holds for  $y$ . A different prime dividing two coordinates divides the third, so  $x, y, z$  are pairwise coprime.

If  $x, y > 0$ , take  $(a, b, c) = (x^3, y^3, 9z^3)$ . If one is negative, move that term to the other side; for example  $x < 0$  gives  $((-x)^3, 9z^3, y^3)$ . Each resulting triple is positive and primitive. Every prime other than three has exponent divisible by three in its product, while the exponent at three is  $2 + 3v_3(z)$ . Thus  $R_0 = 3$ . Only finitely many choices of signs and order give the same positive triple, so infinitely many distinct triples occur. Their heights are unbounded because a bounded height admits only finitely many triples.  $\square$

This family is consistent with Theorem 10.5. It refutes only a bound depending on the cubic residual support alone: the growing primes of cube-divisible exponent still contribute to  $\text{rad}(abc)$ . For moving  $R_0$ , neither the finite-list threshold in Siegel’s theorem nor the field discriminant estimate removes the growing point, denominator, or order-index cost. Controlling that cost uniformly remains a global *abc* problem.

## 11. FORMAL VERIFICATION AND REMAINING OBLIGATIONS

The companion development uses Lean 4.32.0 and the repository’s pinned dependencies. The principal companion modules are listed below. Audit modules only display their types and dependency reports.

| Module (within IUTThreeClosures)               | Checked mathematical scope                                                                                                   |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| SmoothLowOmegaCounting                         | Actual prime-power encoding, finite cardinality bounds, and the finite first moment.                                         |
| SignedPrimeSupport                             | Actual prime-exponent sums and the coupled equivalence with the original target.                                             |
| SignedEndpointDyadicObstruction                | The genuine dyadic triples and failure of a separate-endpoint slope below one.                                               |
| ReachableTensorLatticeCriterion                | Unit-determinant span certificate, symmetric span and its missing off-diagonal tensor.                                       |
| FreyPelleEffectiveResidual<br>Corridor20260830 | The integral elliptic identity, integer square-factor descent, explicit scalar implications, and the Frey denominator bound. |
| AnalyticAmplification<br>Continuation20260830  | CRT uniqueness and finite counts; actual radical inequalities; conic and height identities.                                  |
| DVRReachableHaar20260830                       | A minimum determinant from actual columns, the exact quotient cardinality, closed span, and full-rank Haar and log-volume.   |
| GeometryUniformity<br>Continuation20260830     | Norm and ratio identities, logarithmic absorption, explicit corridor implications, and the primitive Frey–Mordell point.     |

| Module (within IUTThreeClosures)             | Checked mathematical scope                                                                                                                                            |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IUTReachabilityContinuation20260830          | Tensor units, independent scaled vectors, and integral tensor-span containment in a coefficient-module hull.                                                          |
| MatveevPellFinitePacket20260830              | The strict Pell approximation, normalized-coefficient algebra, absolute-index absorption, and the final height implication with external bounds stated as hypotheses. |
| IUTTargetReset20260830                       | Covariance of an all-isomorphism collation with repeated labels, its span, and source reindexing under an explicit compatibility identity.                            |
| AnalyticActualRadical<br>UniformGate20260830 | Actual new-prime radical and integer excess identities, the conic budget, exponent-window algebra, and exact numerical counterexamples.                               |
| GeometryGlobalUniformGate20260830            | Canonical integer cube decomposition, auxiliary point and cubic-root identities, finite matrix determinants, and explicit algebraic height bridges.                   |
| IUTAdmissibleGalois<br>UniformGate20260830   | Kernel and valuation consequences of an explicit trace-transport hypothesis, affine-depth separation, and invariance of module span under unit scalars.               |
| IUTTracePreserving<br>Transvection20260830   | Actual linear automorphisms achieving common avoidance while preserving a functional.                                                                                 |
| IUTAllOpenLatticeRigidity20260830            | Scalar rigidity from all shrinking line neighborhoods; an actual $\mathbb{Z}_p$ specialization, with surjectivity forcing a unit scalar.                              |

These modules introduce no new mathematical axiom. The validation record lists their exact commands, results, and kernel dependency reports. The pre-existing signed-layer module required an explicit import, closure of an elementary algebra branch, and the correct order of unfolding before rewriting; these were repaired without changing its statements.

The formal scope is deliberately narrower than the full paper. Theorem 2.3 and Corollary 2.4 use Younis and classical smooth-number asymptotics in their mathematical proofs; those external analytic theorems are not formalized here. The full-rank case of Theorem 6.1, including its actual index and Haar measure, is now checked in its normalized-probability formulation on the compact integral lattice. Identifying this with the restriction of ambient local-field Haar measure and the rank-deficient measure-zero case remain paper proofs. The ellipse count and asymptotic amplification estimates, source-specific Kummer and local-field identifications, general independent-action theorem, Matveev’s theorem, (35), the small-generator approximation inputs, and the endomorphism-algebra argument retain their stated paper-proof boundary. In particular, the complete effective finiteness theorem is mathematically unconditional by the cited published theorems, but it has not been formalized as an unconditional Lean theorem. A Lean implication with an explicit approximation or height hypothesis is not a Lean proof of that hypothesis. The new radical count uses the cited  $S$ -unit theorem and an analytic divisor sum outside Lean. The actual Ism classification, local class field theory, Galois cohomology, root-logarithm calculation, and categorical representative reconstruction are also paper proofs. The shrinking-line rigidity theorem proves their algebraic core without assuming LCFT as an axiom.



The global cubic construction does not formalize Pasten’s theorem, number-field orders, Siegel’s local approximation, or Nagell–Lutz merely by checking their coordinate identities. These distinctions apply equally to the new module-hull and fixed-support balance statements.

The surviving proof target is a uniform estimate for the coupled expression (25), or a source-derived theorem strong enough to imply it. The IUT route still needs the comparison of the actual output family with the region in the global upper estimate, under compatible normalizations. The analytic disproof route still needs a height-unbounded family satisfying one fixed radical exponent gap; the finite-support bound neither constructs nor forbids such a sparse family. The specified Pell–Chebyshev residual is now effectively finite; the fundamental-unit cases and the absence of a global *abc* reduction remain outside that conclusion. The actual-radical amplifier still requires a lower count in the surviving window; the cubic route must control the growing omitted endpoint or denominators uniformly; the local IUT route still needs the actual Ind1 image and the subsequent Ind3/global comparison. Eliminating Ind2 from one fixed *B*-module hull does not eliminate those tasks. No unconditional closed term of type **ABCConjecture**, or of its negation, is produced by this work.

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